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Pump systems with positioning features

Abstract

Pump systems are described. An example pump system may include a processing unit and a recess configured to receive a pump cartridge. The recess can include a plurality of cartridge engagement slots. The cartridge engagement slots can be configured for removably coupling the pump cartridge. The recess can also include a cartridge-facing surface with a positioning feature slot extending orthogonal to a general plane of the cartridge-facing surface.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4236880	12/1979	Archibald	92/92	F04B 43/0054
4842584	12/1988	Pastrone	N/A	N/A
5039279	12/1990	Natwick	417/474	A61M 5/14228
5098262	12/1991	Wecker et al.	N/A	N/A
5464391	12/1994	DeVale	604/118	A61M 3/022
5554013	12/1995	Owens et al.	N/A	N/A
5575632	12/1995	Morris	N/A	N/A
5603613	12/1996	Butterfield et al.	N/A	N/A
5807075	12/1997	Jacobsen et al.	N/A	N/A
5816779	12/1997	Lawless et al.	N/A	N/A
5954485	12/1998	Johnson et al.	N/A	N/A
6475178	12/2001	Krajewski et al.	N/A	N/A
6494694	12/2001	Lawless et al.	N/A	N/A
6942473	12/2004	Abrahamson	417/474	A61M 5/14228
7867189	12/2010	Childers et al.	N/A	N/A
7972306	12/2010	Shearn	N/A	N/A
8066671	12/2010	Busby et al.	N/A	N/A
8465454	12/2012	Kirkpatrick	N/A	N/A
8523816	12/2012	Kirkpatrick	N/A	N/A
8668671	12/2013	Kirkpatrick	N/A	N/A
8771228	12/2013	Butterfield	N/A	N/A
8784359	12/2013	Plahey et al.	N/A	N/A
8936447	12/2014	Abal	N/A	N/A
2001/0051789	12/2000	Parsons	N/A	N/A
2004/0019313	12/2003	Childers et al.	N/A	N/A
2007/0213653	12/2006	Childers et al.	N/A	N/A
2008/0138223	12/2007	Lanigan et al.	N/A	N/A
2008/0262409	12/2007	Derrico et al.	N/A	N/A
2009/0062738	12/2008	Ziegler	N/A	N/A
2010/0241062	12/2009	Morris	604/29	A61M 1/155
2010/0286599	12/2009	Ziegler et al.	N/A	N/A
2011/0028937	12/2010	Powers	N/A	N/A
2011/0040244	12/2010	Busby et al.	N/A	N/A

2011/0092894	12/2010	McGill et al.	N/A	N/A
2011/0178359	12/2010	Hirschman et al.	N/A	N/A
2011/0282276	12/2010	Abal	N/A	N/A
2012/0053557	12/2011	Abal	N/A	N/A
2012/0078218	12/2011	Barnes	N/A	N/A
2012/0083759	12/2011	Kirkpatrick	N/A	N/A
2012/0177543	12/2011	Battrell et al.	N/A	N/A
2012/0179130	12/2011	Barnes et al.	N/A	N/A
2013/0106609	12/2012	Singh et al.	N/A	N/A
2013/0267899	12/2012	Robert et al.	N/A	N/A
2014/0276424	12/2013	Davis et al.	N/A	N/A
2014/0276426	12/2013	Borges et al.	N/A	N/A
2014/0276533	12/2013	Butterfield et al.	N/A	N/A
2016/0151561	12/2015	Toro et al.	N/A	N/A
2017/0032152	12/2016	Salem et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
WO-2005097235	12/2004	WO	N/A
WO-2014190188	12/2013	WO	N/A
WO-2016190904	12/2015	WO	N/A

OTHER PUBLICATIONS

Extended European Search Report for Application No. 15864547.3, dated Aug. 1, 2018, 7 pages. cited by applicant

Extended European Search Report for Application No. 15865327.9, dated Aug. 1, 2018, 7 pages. cited by applicant

International Search Report and Written Opinion for Application No. PCT/US2015/063001, dated Mar. 8, 2016, 22 pages. cited by applicant

International Search Report and Written Opinion for Application No. PCT/US2015/063002, dated Mar. 8, 2016, 11 pages. cited by applicant

International Search Report and Written Opinion for Application No. PCT/US2015/063007, dated Mar. 8, 2016, 10 pages. cited by applicant

International Search Report and Written Opinion for Application No. PCT/US2015/063010, dated Mar. 8, 2016, 13 pages. cited by applicant

International Search Report and Written Opinion for Application No. PCT/US2015/063013, dated Mar. 8, 2016, 15 pages. cited by applicant

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional application of U.S. application Ser. No. 16/365,541, filed Mar. 26, 2019, issued as U.S. Pat. No. 11,752,255 on Sep. 12, 2023, which is a divisional application of U.S. application Ser. No. 14/557,447, filed Dec. 1, 2014, issued as U.S. Pat. No. 10,245,373 on Apr. 2, 2019, the disclosures of which are herein expressly incorporated by reference in their entirety.

TECHNICAL FIELD

(1) The present disclosure generally relates to apparatus, systems, and methods of delivering medical fluid to patients, and more particularly to infusion pumps, disposable cassettes, and associated methods.

BACKGROUND

(2) Infusion pumps are medical devices that may be used to administer intravenous (IV) fluids. An infusion pump can facilitate the delivery of IV fluids while controlling the volumes and rates for the delivery of such IV fluids. The IV fluids may be delivered at continuous rates or intermittent intervals. Some infusion pumps move fluid through an IV tube using a peristaltic pumping mechanism that acts on the IV tube, while other infusion pumps rely on a cartridge or cassette-like device intended to be manipulated by a pump to cause the IV fluid to flow at the controlled rate or interval. In either case, a typical infusion pump, manipulates the IV tube or IV cartridge such that the IV fluid moves from a container to a patient. The IV tube or IV cartridge is typically connected to or integrated with an IV set (e.g., tubing, valves, and fittings for delivering fluid to a patient), and therefore the cartridge and IV set may be disposable to reduce the risk of infection and contamination.

SUMMARY

(3) Aspects of the subject technology relate to pump systems. In accordance with certain aspects, a pump system may comprise a processing unit; and a recess configured to receive a pump cartridge, the recess comprising: a plurality of cartridge engagement slots configured for removably coupling the pump cartridge; and a cartridge-facing surface comprising a positioning feature slot extending orthogonal to a general plane of the cartridge-facing surface.

(4) In accordance with certain aspects, a pump system may comprise a processing unit; and a cartridge recess configured to receive a pump cartridge, the cartridge recess comprising: a cartridge-facing surface; and a plurality of cartridge engagement slots configured for removably coupling the pump cartridge, wherein each of the plurality of cartridge engagement slots comprises a flat face ramp portion.

(5) In accordance with certain aspects, an pump system may comprise a processing unit; a cartridge recess configured to receive a pump cartridge, the cartridge recess comprising: a cartridge-facing surface including a positioning feature slot extending orthogonal to a general plane of the cartridge-facing surface, wherein the positioning feature slot comprises: a first positioning feature slot operably coupled to a pump actuator configured to apply a force orthogonal to a general plane of cartridge-facing surface; and a second positioning feature slot operably coupled to a fluid sensor.

(6) It is understood that in accordance with certain aspects, the cassette recess may be integrated into the same box as the processing unit or may be contained in an interface module that may be operatively coupled to the processing unit.

(7) It is understood that various configurations of the subject technology will become readily apparent to those skilled in the art from the disclosure, wherein various configurations of the subject technology are shown and described by way of illustration. As will be realized, the subject technology is capable of other and different configurations and its several details are capable of modification in various other respects, all without departing from the scope of the subject technology. Accordingly, the summary, drawings and detailed description are to be regarded as illustrative in nature and not as restrictive.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are included to provide further understanding and are incorporated in and constitute a part of this specification, illustrate disclosed embodiments and

together with the description serve to explain the principles of the disclosed embodiments. In the drawings:

- (2) FIGS. 1A and 1B are overview diagrams illustrating examples of infusion pump systems, in accordance with aspects of the present disclosure.
- (3) FIGS. 2A and 2B illustrate perspective views of examples of a first embodiment disposable IV pump cassette and cassette recess, in accordance with aspects of the present disclosure.
- (4) FIG. 2C illustrates a front perspective view of an example of a first embodiment disposable IV pump cassette, in accordance with aspects of the present disclosure.
- (5) FIG. 2D illustrates a side perspective view of an example of a first embodiment disposable IV pump cassette, in accordance with aspects of the present disclosure.
- (6) FIG. 2E illustrates a rear perspective view of an example of a variation of a first embodiment disposable IV pump cassette including a detail section (Detail A), in accordance with aspects of the present disclosure.
- (7) FIG. 2F illustrates a rear perspective view of an example of a variation a first embodiment disposable IV pump cassette, in accordance with aspects of the present disclosure.
- (8) FIG. 3A is an exploded perspective detail view illustrating an example of a first embodiment disposable IV pump cassette, in accordance with aspects of the present disclosure.
- (9) FIG. 3B illustrates a perspective view of an example of a first embodiment disposable IV pump cassette, in accordance with aspects of the present disclosure.
- (10) FIG. 3C illustrates a perspective view of an example of a first embodiment cassette recess, in accordance with aspects of the present disclosure.
- (11) FIG. 3D illustrates an enlarged view of an example of a pump chamber area of a first embodiment disposable IV pump cassette, in accordance with aspects of the present disclosure.
- (12) FIG. 4A illustrates perspective views of examples of a first embodiment disposable IV pump cassette and cassette recess, in accordance with aspects of the present disclosure.
- (13) FIG. 4B illustrates perspective views of examples of a first embodiment disposable IV pump cassette and cassette recess, in accordance with aspects of the present disclosure.
- (14) FIG. 4C illustrates perspective views of examples of a first embodiment disposable IV pump cassette and cassette recess, in accordance with aspects of the present disclosure.
- (15) FIG. 5A illustrates an enlarged perspective view of an example of a first embodiment disposable IV pump cassette, in accordance with aspects of the present disclosure.
- (16) FIG. 5B illustrates an enlarged perspective view of an example of a first embodiment disposable IV pump cassette, in accordance with aspects of the present disclosure.
- (17) FIG. 5C illustrates a cross-sectional view of an example of a portion of a first embodiment disposable IV pump cassette, in accordance with aspects of the present disclosure.
- (18) FIG. 5D illustrates a perspective view of an example of a portion of a first embodiment disposable IV pump cassette, in accordance with aspects of the present disclosure.
- (19) FIG. 5E illustrates a perspective view of an example of a portion of a first embodiment disposable IV pump cassette, in accordance with aspects of the present disclosure.
- (20) FIG. 5F illustrates a view of an example of a portion of a first embodiment disposable IV pump cassette, in accordance with aspects of the present disclosure.
- (21) FIG. 6A illustrates perspective views of examples of second embodiment disposable IV pump cassette and cassette recess, in accordance with aspects of the present disclosure.
- (22) FIG. 6B is an exploded detail view illustrates a perspective view of second embodiment disposable IV pump cassette, in accordance with aspects of the present disclosure.
- (23) FIGS. 7A and 7B illustrate perspective views of examples of a third embodiment disposable IV pump cassette and cassette recess, in accordance with aspects of the present disclosure.
- (24) FIG. 7C illustrates a front perspective view of an example of a third embodiment disposable IV pump cassette, in accordance with aspects of the present disclosure.
- (25) FIG. 7D illustrates a side perspective view of an example of a third embodiment disposable IV

pump cassette, in accordance with aspects of the present disclosure.

(26) FIGS. **8A** and **8B** are exploded perspective detail views illustrating an example of a third embodiment disposable IV pump cassette, in accordance with aspects of the present disclosure.

(27) FIG. **8C** illustrates a perspective view of an example of a third embodiment disposable IV pump cassette, in accordance with aspects of the present disclosure.

(28) FIG. **8D** illustrates a perspective view of an example of a third embodiment cassette recess, in accordance with aspects of the present disclosure.

(29) FIG. **8E** illustrates an enlarged cross-sectional view of an example of a pump chamber area of a third embodiment disposable IV pump cassette, in accordance with aspects of the present disclosure.

(30) FIG. **8F** illustrates an enlarged cross-sectional view of an example of a flow stop valve portion of a third embodiment disposable IV pump cassette, in accordance with aspects of the present disclosure.

(31) FIG. **9A** illustrates perspective views of examples of a third embodiment disposable IV pump cassette and cassette recess, in accordance with aspects of the present disclosure.

(32) FIG. **9B** illustrates side perspective views of examples of a third embodiment disposable IV pump cassette and cassette recess, along with examples of interface structures, in accordance with aspects of the present disclosure.

(33) FIG. **10A** illustrates perspective views of examples of a fourth embodiment disposable IV pump cassette and cassette recess, in accordance with aspects of the present disclosure.

(34) FIG. **10B** illustrates perspective views of examples of a fourth embodiment disposable IV pump cassette and cassette recess, in accordance with aspects of the present disclosure.

(35) FIG. **10C** illustrates perspective views of examples of a fourth embodiment disposable IV pump cassette and cassette recess, in accordance with aspects of the present disclosure.

(36) FIG. **11A** illustrates perspective views of an example of a fourth embodiment disposable IV pump cassette, in accordance with aspects of the present disclosure.

(37) FIG. **11B** illustrates perspective views of an example of a fourth embodiment disposable IV pump cassette, in accordance with aspects of the present disclosure.

(38) FIGS. **12A** and **12B** illustrate perspective views of examples of a fifth embodiment disposable IV pump cassette and cassette recess, in accordance with aspects of the present disclosure.

(39) FIG. **12C** illustrates a front perspective view of an example of a fifth embodiment disposable IV pump cassette, in accordance with aspects of the present disclosure.

(40) FIG. **12D** illustrates a side perspective view of an example of a fifth embodiment disposable IV pump cassette, in accordance with aspects of the present disclosure.

(41) FIG. **12E** illustrates an enlarged cross-sectional perspective view of an example of a fifth embodiment disposable IV pump cassette, in accordance with aspects of the present disclosure.

(42) FIG. **13A** illustrates a perspective view of an example of a fifth embodiment disposable IV pump cassette, in accordance with aspects of the present disclosure.

(43) FIG. **13B** illustrates a perspective view of an example of a fifth embodiment disposable IV pump cassette, in accordance with aspects of the present disclosure.

(44) FIG. **13C** illustrates an enlarged cross-sectional perspective view of an example of a fifth embodiment disposable IV pump cassette, in accordance with aspects of the present disclosure.

(45) FIG. **13D** illustrates an enlarged cross-sectional perspective view of an example of a fifth embodiment disposable IV pump cassette, in accordance with aspects of the present disclosure.

(46) FIG. **13E** illustrates an enlarged cross-sectional view of an example of a fifth embodiment disposable IV pump cassette, in accordance with aspects of the present disclosure.

(47) FIG. **13F** illustrates a perspective view of an example of a cassette recess, in accordance with aspects of the present disclosure.

(48) FIG. **14A** illustrates perspective views of examples of a fifth embodiment disposable IV pump cassette and cassette recess, in accordance with aspects of the present disclosure.

- (49) FIG. 14B illustrates perspective views of examples of a fifth embodiment disposable IV pump cassette and cassette recess, in accordance with aspects of the present disclosure.
- (50) FIG. 14C illustrates perspective views of examples of a fifth embodiment disposable IV pump cassette and cassette recess, in accordance with aspects of the present disclosure.
- (51) FIG. 14D illustrates perspective views of examples of a fifth embodiment disposable IV pump cassette and cassette recess, in accordance with aspects of the present disclosure.
- (52) FIG. 14E illustrates rear perspective views of examples of a fifth embodiment disposable IV pump cassette and cassette recess, along with examples of interface structures, in accordance with aspects of the present disclosure in accordance with aspects of the present disclosure.
- (53) FIG. 15A illustrates perspective views of examples of a sixth embodiment disposable IV pump cassette and cassette recess, in accordance with aspects of the present disclosure.
- (54) FIG. 15B illustrates an enlarged cross-sectional perspective view of an example of a sixth embodiment disposable IV pump cassette, in accordance with aspects of the present disclosure.
- (55) FIG. 15C illustrates a perspective view of an example of a sixth embodiment disposable IV pump cassette, in accordance with aspects of the present disclosure.

DETAILED DESCRIPTION

(56) The detailed description set forth below describes various configurations of the subject technology and is not intended to represent the only configurations in which the subject technology may be practiced. The detailed description includes specific details for the purpose of providing a thorough understanding of the subject technology. Accordingly, dimensions may be provided in regard to certain aspects as non-limiting examples. However, it will be apparent to those skilled in the art that the subject technology may be practiced without these specific details. In some instances, well-known structures and components are shown in block diagram form in order to avoid obscuring the concepts of the subject technology.

(57) It is to be understood that the present disclosure includes examples of the subject technology and does not limit the scope of the appended claims. Various aspects of the subject technology will now be disclosed according to particular but non-limiting examples. Various embodiments described in the present disclosure may be carried out in different ways and variations, and in accordance with a desired application or implementation.

(58) FIG. 1A illustrates an example of an infusion pump system. In accordance with certain embodiments, infusion pump system **10** may include one or more cassette recesses and disposable IV pump cassettes (e.g., cassette recesses **200**, **400**, **600** and cassettes **100**, **300**, and **500**). For example, cassette recess **200** may be configured to receive cassette **100** and provide various mechanical couplings and operational interfaces (e.g., fittings, motor, gearing, driveshaft, sensors, etc.). Infusion pump system **10** may include central processing unit **12** with display screen **14** (e.g., touchscreen display), and a data input features **16**, for example, a keypad and a series of configurable buttons adjacent to display screen **14**. Other types of input and output devices may be used with central processing unit **12** and infusion pump system **10**. In certain aspects, central processing unit **12** is operatively coupled to one or more interface modules, with cassette recesses **200**, to control and communicate with various operational interfaces thereof.

(59) FIG. 1B illustrates another example of an infusion pump system. In accordance with certain embodiments, infusion pump system **11** may include one or more cassette recesses and disposable IV pump cassettes (e.g., cassette recesses **200**, **400**, **600** and cassettes **100**, **300**, and **500**). For example, cassette recess **200** may be configured to receive cassette **100** and provide various mechanical couplings and operational interfaces (e.g., fittings, motor, gearing, driveshaft, sensors, etc.). Infusion pump system **11** may include central processing unit **13** with display screen **15** (e.g., touchscreen display), and a data input features **17**, for example, a series of configurable buttons adjacent to display screen **15**. In some implementations, the display screen **15** may provide a keypad or similar data entry feature. Other types of input and output devices may be used with central processing unit **13** and infusion pump system **11**. In certain aspects, central processing unit

13 is operatively coupled to one or more interface modules, with cassette recesses **200**, to control and communicate with various operational interfaces thereof.

(60) In operation, an IV bag, syringe or other fluid source **52** may be fluidly connected to inlet **112** of cassette **100**, and outlet **114** of cassette **100** may be fluidly connected to a patient **54** as shown in the examples of FIGS. **1A** and **1B**. Infusion pump systems **10** and **11** may be configured to operate over a wide range of infusion rates such as, but not limited to, 1-999 ml/h for general purpose and operating room applications, and 0.1-99.9 ml/h for neonatal applications. Infusion pump systems **10** and **11** may include, for example, low-sorbing configurations compatible with chemotherapy, TPN and Nitroglycerin (NTG). In accordance with some embodiments, cassettes **100** may comprise a DEHP-free and Latex fluid pathway suitable for various patient populations (e.g., neonate, pediatric, and adult).

(61) In accordance with aspects of the subject technology, disposable IV cassettes **100** used with infusion pump systems **10** and **11** may be substantially reduced in size when compared to conventional disposable IV cassette units resulting in a significant amount of medical plastic required to be treated and disposed of in compliance with various regulations.

(62) Additionally, infusion pump systems **10** and **11** comprising externally mounted and translucent cassettes **100** for which fluid passage through the entire fluid pathway, or a portion thereof, in the IV set may be advantageous.

(63) In operation, a user (e.g., a caregiver) may obtain a new disposable IV cassette **100** and prime cassette **100** before inserting cassette **100** into cassette recess **200**. Caregiver may check for any visible air bubbles in the fluid pathway and may press on any accessible fluid reservoirs (e.g., pressure dome chambers) to move fluid through the cassette **100**. In accordance with certain aspects, cassette **100** can be securely held and inserted into cassette recess **200** by a single hand of a caregiver. In this regard, caregiver's other hand can be freed to perform other tasks.

(64) FIGS. **2A** and **2B** illustrate examples of a disposable IV pump cassette **100** and corresponding cassette recess **200** of an interface module. In accordance with certain embodiments, cassette **100** may comprise a cassette body **110** and a slider **170**. Cassette **100** may include certain may include certain visual indicators related to operation aspects of the cassette and the infusion pump system in general. For example, cassette may include identifiable images such as fluid drops indicating position of slider **170** for free-flow (flow stop valve **164** in an open position) and a patient figure proximal to outlet **114**. In accordance with some aspects, cassette **100** may include lens area **173** for magnification of the fluid pathway within the cassette body **110**. Lens area **173** may be disposed on the slider **170** or proximal to outlet **114** and/or an air-in-line detection feature. For example, during priming or prepping a cassette, a user or caregiver may use lens area **173** to ensure that any visible air bubbles have been removed and fluid is flowing properly. In accordance with some aspects, one or more cassette-seated sensors may be disposed within the cassette recess **200** so as to inform central processing unit **12** that the cassette is locked or secured into place within the cassette recess **200** or seat.

(65) Slider **170** can be fixably and slidably engaged with cassette body **110** such that slider **170** may articulate longitudinally **191** with respect to cassette body **110**, but will be constrained within range of sliding motion such that the slider remains coupled to the cassette body **110**. Slider **170** may be formed from rigid plastic or polymer material having lubricating characteristics (e.g., incorporating silicon or polytetrafluoroethylene (PTFE) additives), and is clear or translucent in accordance with certain embodiments. In some embodiments, slider **170** may be polycarbonate. Slider **170** includes a slider grip **172** or handle portion and a plurality of protrusions **174** or lugs that are configured to be releasably lockable with a plurality of slots **274** of the cassette recess **200** (e.g., L-shaped locking channels). In this regard, cassette **100** can be self-latched into the cassette recess **200**. Accordingly, a door or lever action is not required in order to retain the cassette **100** within the cassette recess **200**. In an alternative embodiment, an inverse configuration may be desired, in which the cassette recess **200** would contain protrusions or lugs that would be

configured to be releasably lockable with a corresponding slots located on the slider or rigid body. (66) The plurality of protrusions **174** may be disposed at various locations on slider **170**. As illustrated in the example of FIG. 2C, slider **170** may comprise a first pair of protrusions **174a** and a second pair of protrusions **174b**. The first pair of protrusions **174a** may be spaced apart a different distance (e.g., width **W1**) than the second pair of protrusions **174b** (e.g., width **W2**). For example, in certain aspects, **W1** may be between 31 mm and 35 mm, and **W2** may be between 21 mm and 25 mm. In this regard, cassette **100** orientation within cassette recess **200** (along with non-vertically aligned inlet recess **212** and outlet recess **214**) is clear to a user or caregiver such that a cassette **100** is not inadvertently installed (or when being primed) in an inverted manner.

(67) Additionally, perimeter edges **152** of cassette **100** and corresponding perimeter edges of cassette recess **200** may include a plurality of arcuate perimeter edges for engagement between cassette **100** and cassette recess **200**. For example, the plurality of arcuate perimeter edges may correspond to a perimeter opening of cassette recess **200** for front-loading cassette **100** into cassette recess **200**. With reference to FIGS. 2C and 2D, cassette body **110** (with or without slider **170**) may comprise top lateral edge **152a**, straight longitudinal edge **152b**, arcuate side edge **152c**, arcuate side edge **152d**, straight longitudinal edge **152e**, bottom arcuate edge **152f**, bottom arcuate edge **152g**, and straight longitudinal edge **152h**. Straight longitudinal edge **152b**, straight longitudinal edge **152e**, and straight longitudinal edge **152h** may all be parallel and/or opposing edges to each other, in accordance with certain example aspects.

(68) Arcuate side edge **152c**, bottom arcuate edge **152f**, and bottom arcuate edge **152g** may be concave curved edges with respect to a general center portion of cassette **100** (e.g., pump chamber, valve, and positioning features), whereas arcuate side edge **152d** is a convex curved edge with respect to a general center portion of cassette **100**. Additionally, cassette **100** may include corner edges where two straight edges converge. For example, corner edges may be defined where top lateral edge **152a** and straight longitudinal edge **152b** meet, as well as where top lateral edge **152a** and straight longitudinal edge **152h** meet. Corner edges may be slightly rounded at the point of intersection in some implementations, but not considered arcuate with respect to defining an arcuate edge portion or section of cassette **100** (or cassette recess **200**) for front-loading features described herein. Moreover, the arcuate edges on cassette **100** may have different curvatures, for example, different arc lengths, gradients, etc., such that an arcuate edge may be distinguishable from other arcuate edges curved edges as well as rounded corner edges. In other embodiments, perimeter edges **152** of cassette **100** may include a configuration of straight-line edges. In still other embodiments, one or more of the corner edges may be rounded sufficiently to be considered arcuate with respect to defining an arcuate edge portion or section of the cassette **100** (or cassette recess **200**) for front-loading features described herein.

(69) Without being limited to the specific embodiments described above, generally the arcuate edges of the cassette **100** provide a benefit by enabling visual identification as well as prohibiting improper orientation or seating of cassette **100** with a corresponding cassette recess **200**.

Additionally, minimizing a size of cassette **100** and optimizing a surface area of one or more corresponding cassette recess **200** may be achieved using arcuate perimeter edge configurations.

(70) Additionally, an overall size of cassette **100** and cassette recess **200** may be reduced, in accordance with some aspects. For example, in certain embodiments, cassette body **110** may extended longitudinally a length (**L**) between 90 mm and 100 mm. In this regard, cassette body **110** may be dimensioned as having a first body section of a first width and a second body section of a second width. The first body section may have a wider width than the second body section, and the second body section may have a longer length than the first body section. For orientation reference with respect to the various views of the examples illustrated of FIGS. 2C-2F, longitudinal axis or y-axis **195**, latitudinal axis or x-axis **196**, and depth axis or z-axis **197** are provided.

(71) In this regard, depth aspects of cassette **100** is shown in the example of FIG. 2D. For example, in certain embodiments, cassette body **110**, or a substantial portion thereof, may extend depth (**D**)

between 6 mm and 8 mm. Fluid pathway extension member **128** may further extend between 8 mm to 10 mm. In certain aspects, slider grip **172** may extend between 10 mm to 14 mm from cassette body **110**. It is to be appreciated that the process of cleaning of inlet recess **212**, outlet recess **214**, and cassette recess **200** is made efficient in the shallow recess configuration in accordance with certain embodiments should any fluid or debris accumulate within cassette recess **200**. The shallow recess configuration of cassette recess **200**, and associated longitudinal alignment of cassette **100** such that a smaller of volumetric dimensions of cassette **100** (e.g., depth being smaller than length and width in certain embodiments) further enables additional space for arrangement of mechanical couplings and operational interfaces and optimizes the overall space requirements of cassette recess **200** and infusion pump system in general.

(72) Various types, placement, and orientations of the plurality of protrusions **174** disposed on slider **170** are contemplated in the present disclosure. As illustrated in the example of FIG. 2E, plurality of protrusions **174c** can protrude toward the cassette recess **200** (e.g., along z-axis **197**), and be receivable by corresponding slots **274c** configured to receive protrusion **174c**. As shown in Detail A of FIG. 2E, protrusion **174c** can be aligned and inserted into an opening of slot **274c** having a circular cross-section with a lower slit of reduced diameter than the circular cross-section (e.g., keyhole latch design). Slider **170** may then be articulated downwardly when a head portion of protrusion **174c** is aligned with an internal longitudinal channel of slot **274c** and a neck portion of protrusion **174c** is aligned with the lower slit of slot **274c**. The present disclosure contemplates protrusions **174** of any size or shape on the outward edges of slider **170** so long as the corresponding slots **274** are of matching dimensions and can serve to latch or cassette **100** in the cassette recess **200** through the movement of the slider **170** from an disengaged to engaged position.

(73) As illustrated in the example of FIG. 2F, cassette **100** may be configured with one or more protrusions **113** disposed on cassette body **110** and a plurality of protrusions **174** disposed on slider **170**. The one or more cassette body protrusions **113** may engage with one or more corresponding upper slots **274** independently from the plurality of protrusions **174**. In this regard, cassette **100** may be configured to hang or otherwise be securely positioned in cassette recess **200**, without operatively engaging associated mechanical couplings and operational interfaces of cassette recess **200**. When cassette **100** is to become operational, cassette **100** may be pivoted (or removed and reinserted), for example, such that protrusions **174** securely engage with the lower slots **274** of cassette recess **200**.

(74) In accordance with other embodiments, cassette **100** may be coupled to cassette recess **200** by the one or more protrusions **113** and not necessarily comprise a slider mechanism. For example, a grip section may be disposed on cassette body **110** and the one or more protrusions **113** may be configured to engage with and securely latch to one or more slots **274** of the cassette recess **200**.

(75) In other embodiments, a single protrusion or latch disposed on one of cassette **100** or slider **170** may be utilized to couple with cassette recess **200**. Aspects of the various cassette-coupling techniques illustrated in the examples of FIGS. 2C-2F and described herein may be further combined and arranged into additional configurations suitable for specific implementations given the benefit of the present disclosure.

(76) Moreover, in accordance with certain aspects, features of cassette recess **200** are designed to avoid wear down and/or risk of malfunction. For example, the plurality of slots **274** arranged within cassette recess **200** may be devoid of any movable latching mechanism in certain embodiments as such movable latching mechanisms may be susceptible to excessive wear and mechanical failure over repeated use with multiple disposable IV cassettes **100**.

(77) In operation, cassette **100** can be loaded directly into cassette recess **200**. In this regard, the direct loading of the cassette **100** will enable avoidance of sheer forces that might otherwise be applied to the sensors, alignment features, and other engaging interfaces of cassette-facing surface **216** of cassette recess **200** from interaction with the interface-facing side of cassette body **110** as it

is loaded into cassette recess **200**.

(78) It is to be understood that modification to the various features of cassette **100** can be made to accommodate the various cassette-coupling techniques disclosed herein.

(79) Referring now to the examples of FIGS. **3A** and **3B**, cassette body **110** may comprise interface-facing frame portion **116** and slider-facing base portion **119** with membrane **117** disposed substantially therebetween (e.g., portions of membrane **117** may extend through some openings of frame portion **116**). In accordance with certain embodiments, membrane **117** can be a compliant material co-molded to the frame portion **116** and sealingly engaged with base portion **119** for defining a fluid pathway through cassette body **110** from inlet **112** to outlet **114**. Mating edges of frame portion **116** and base portion **119** may be connected by fusing, welding, gluing, or the like. Membrane **117** and base portion **119** may further define a plurality of other features, some of which may be accessed through openings in frame portion **116**.

(80) Frame portion **116**, membrane **117**, and/or base portion **119** may define features in or along the fluid pathway, in accordance with certain embodiments. For example, beginning from inlet **112**, the fluid pathway may include features such as, but not limited to, upstream pressure dome **132** (e.g., an inlet-side compliant reservoir), inlet-side valve **122**, pump chamber having pump chamber opening/access **125**, outlet-side valve **124**, downstream pressure dome **134** (e.g., an outlet-side compliant reservoir), fluid pathway extension member **128**, and flow stop valve **164**. Other features that are not in or along the fluid pathway, but are disposed on cassette body **110**, may include positioning port **120** and slider stopper **151**. With respect to extension member **128**, a portion of the fluid pathway can be extended away or protrude orthogonally from the generally flat and planar exterior surface of interface-facing frame portion **116** so as to make the fluid in the fluid pathway available for certain detection techniques performed by infusion pump system **10**, **11**. As illustrated in the example of FIGS. **3A** and **3B**, fluid pathway extension member **128** may be formed from orthogonally extending portions of frame portion **116**, membrane **117**, and/or base portion **119**.

(81) In accordance with certain embodiments, membrane **117** may be formed from a thermoplastic elastomer (TPE). Characteristics of certain TPEs can enable effective co-molding with other materials, for example, polycarbonate. Accordingly, in some embodiments, membrane **117** may be co-molded to frame portion **116** and striker **181** may be co-molded to a portion of membrane **117** defining a flow stop valve **164**. However, in some embodiments, membrane **117** can be formed from silicon, a silicon-based compound, an elastomeric material suitably compliant for fluid flow, or the like.

(82) In accordance with certain embodiments, interface-facing frame portion **116** and slider-facing base portion **119** may be formed from a rigid plastic such as, but not limited, a polycarbonate. Additionally, the rigid plastic of frame portion **116** and base portion **119** may be clear or translucent. The material of membrane **117** (e.g., TPE or other compliant material) and rigid plastic slider **170** may also be clear or translucent, thereby allowing a user or caregiver to readily observe fluid passage through a substantial portion of the fluid pathway of cassette body **110**. In some embodiments, the fluid pathway portion of cassette body **110** will be clear or translucent, and other portions will be frosted so as to direct a user or caregiver's attention to the fluid pathway.

(83) In some implementations, slider **170**, base portion **119**, and membrane **117** may be clear or translucent (or at least some portions along the fluid pathway), and the frame portion **116** may not be translucent. For example, the frame portion **116** may be colored in a manner so as to contrast against a color or tint of the fluid expected to be used with cassette **100**. In some embodiments, a lens area **173** may be disposed on base portion **119** alternatively, or in addition to, lens area **173** disposed on slider **170**.

(84) With additional reference to the example of FIG. **3C**, one or more fluid sensors may be disposed within sensor slot **228**. The one or more fluid sensors disposed within sensor slot **228** can be ultrasonic sensors configured as an air-in-line detector, for example. In certain embodiments, extension member **128** may be disposed on cassette body **110** and positioned along the fluid

pathway between downstream pressure dome **134** and flow stop valve **164**. However, in some embodiments, extension member **128** can be positioned at other locations along the fluid pathway such as, but not limited to, between inlet **112** and upstream pressure dome **132**. Additionally, in other embodiments, a plurality of extension members **128** with a plurality of corresponding sensor slots **228** may be positioned along a fluid pathway of cassette body **110**.

(85) As illustrated in the examples of FIG. **3A-3C**, cassette body **110** may include a pump drive assembly in accordance with certain embodiments. For example, the pump drive assembly may include pump drive interface **142** for receiving pump actuator **242** of cassette recess **200**. Pump drive interface **142** can be operatively coupled to piston **145** slidably engaged within piston guide **143** or casing (e.g., generally cylindrical or frustoconical casing) such that reciprocal movement of piston **145** moves within pump chamber providing a seal to urge fluid through the fluid pathway of cassette body **110**. In this regard, the pump chamber may be defined by a portion of the piston guide **143** or casing distal from the pump drive interface **142** that is adjacent to and fluidly coupled with a tract or section of the fluid pathway between inlet-side valve **122** and outlet-side valve **124**. Thus, pump chamber permits movement of the piston guide **143** with the reciprocal motion of the piston **145** such that a volume of the pump chamber may be varied by movement of the piston **145** in accordance with certain embodiments. In accordance with certain aspects, piston **145** resides and moves within a rigid bore and allows a seal that permits fluid to be drawn into the pump chamber via pump chamber opening/access **125** on the fill cycle and expelled on the delivery cycle.

(86) A wiper seal (not shown) may be positioned within or proximal to piston guide **143** and slidably engaged with piston **145** thereby reducing the possibility of any substances (e.g., dirt or dried fluid particles) near the cassette **100** from contacting one or more slidable seals of the piston **145**. The one or more slideable seals of the piston **145** may contact an internal wall piston guide **143** to form a movable barrier of the pump chamber. Additionally, piston **145** may include a reduced tip portion for more precise volumetric displacement of fluid into and out of pump chamber through pump chamber opening/access **125**.

(87) For example, pumping operation of infusion pump system **10, 11** when cassette **100** is primed and seated in cassette recess **200** may comprise activating outlet-side valve actuator **224** such that outlet-side valve **124** is closed or sealed while activating inlet-side valve actuator **222** such that inlet-side valve **122** is opened. Opening of inlet-side valve **122** may coincide with or occur shortly before the start of a reverse stroke of piston **145** (e.g., a movement of piston **145** away from pump chamber). Accordingly, fluid can flow from upstream pressure dome **132** to pump chamber. Alternatively, or in addition to, outlet-side valve **124** may comprise a one-way valve mechanism that permits flow of fluid under normal conditions in one direction (from a fluid container to a patient). Additionally, in some alternative embodiments, inlet-side valve **122** may also comprise a one-way valve or choke mechanism permitting flow of fluid in primarily one direction (e.g., from a fluid container to a patient) under normal operating conditions. In this configuration, cassette recess **200** may not need to incorporate either outlet-side valve actuator **224** or inlet-side valve actuator **222**. Outlet-side valve **124** and inlet-side valve **122** may limit flow of fluid in one direction, but permit flow in an opposite direction in the event fluid pressure overcomes a cracking pressure of the valves.

(88) Continuing with the valve-operated implementation, pumping operation may comprise activating outlet-side valve actuator **224** such that outlet-side valve **124** is open while activating inlet-side valve actuator **222** such that inlet-side valve **122** is closed or sealed. Opening of outlet-side valve **124** may coincide with or occur shortly before a start of a forward stroke of piston **145** (e.g., a movement of piston **145** toward the opening/access **125** of the pump chamber such that the volume of the pump chamber is reduced). Thus, fluid can flow from pump chamber down the fluid pathway to outlet **114**.

(89) In certain embodiments, the upstream pressure dome **132** may be smaller than the downstream pressure dome **134** to minimize retained volume. Likewise the downstream pressure dome **134** may

be larger than the upstream pressure dome **132** to improve resolution of fluid pressure thereby allowing for an accurate and precise volume of fluid to be pumped and any upstream or downstream pressures to be accurately measured.

(90) As illustrated in the example of FIG. **3D**, inlet-side valve **122** and outlet-side valve **124** may be positioned 180° away from each other for optimizing valve actuation mechanics. In certain embodiments, piston **145** is aligned below the opening of pump chamber and parallel to a longitudinal axis **25** of the general center of the pump chamber. In an alternate embodiment, piston **145** is aligned above the opening/access **125** of the pump chamber and parallel to a longitudinal axis **25** of the general center of pump chamber. It is to be appreciated that piston pump techniques can provide repeatedly precise positive displacement of fluid in the pump chamber. Moreover, piston pump techniques as described in the present disclosure may be readily adapted to be syringe pump embodiments.

(91) Referring to FIGS. **3A-3C**, pump drive interface **142** and pump actuator **242** may be configured as a reciprocating motion mechanism (e.g., a scotch-yoke configuration) in certain implementations. In such implementations, pump drive interface **142** may include opposing ramp portions for guiding a rotatable pin of pump actuator **242** toward an elongate slot of pump drive interface **142**. For example, the outer edges of the opposing ramp portions may be arranged a distance that will ensure engagement with the rotatable pin of pump actuator **242**. When the rotatable pin contacts one of the ramp portions, the pump drive interface **142** will move to align the elongate slot of pump drive interface **142** with the with the rotatable pin of pump actuator **242**. However, it is to be appreciated that other pump drive assemblies are contemplated with cassette **100** and cassette recess **200** in accordance with the present disclosure.

(92) For example, pump drive assembly may comprise an orthogonal pump drive mechanism whereby a pump actuator of an interface module applies a force orthogonal to a plane of interface-facing frame portion as described with respect to example cassette **500** and cassette recess **600** embodiments described herein.

(93) In certain embodiments, cassette recess **200** may include an upstream pressure sensing probe **232** and downstream pressure sensing probe **234** enabling measurement of in-line pressure and fault isolation to a section of the fluid pathway. For example, upstream pressure sensing probe **232** may operably contact upstream pressure dome **132** through a corresponding opening of interface-facing frame portion **116**. Similarly, downstream pressure sensing probe **234** may operably contact downstream pressure dome **134** through a corresponding opening of frame portion **116**.

(94) With reference to the examples of FIGS. **3B** and **3C**, the x-y positioning of cassette **100** within cassette recess **200** can be constrained by the positioning port **120** and positioning protrusion **220** mating interface, as well as fluid pathway extension member **128** and sensor slot **228** mating interface (e.g., air-in-line detector feature). In this regard, cassette **100** and cassette recess **200** can have two points of contact in the z-axis direction (e.g., an axis through and transverse to a general plane of the interface-facing surface of cassette body **110** of cassette **100** and cassette-facing surface of cassette recess **200**) for interlock alignment of the cassette with respect to the x-y positioning of the interface side of the cassette body **110**.

(95) According to certain aspects, positioning port **120** may be located proximal to inlet-side valve **122** and outlet-side valve **124**, and correspondingly mating positioning protrusion **220** may be located proximal to inlet-side valve actuator **222** and outlet-side valve actuator **224**. In this regard, the cassette **100** may be positioned properly without over constraining the cassette **100**.

(96) In some embodiments, the latching engagement of slider **170** with cassette recess **200** may be less relevant to the x-y positioning in the cassette body **110** for properly aligning features on the interface-facing frame portion **116**.

(97) In other words, the controlling index for proper x-axis and y-axis alignment of the interface-facing frame portion **116** is the positioning port **120** and positioning protrusion **220** mating interface, in certain embodiments. In some embodiments, the positioning port **120** and positioning

protrusion **220** mating interface, and fluid pathway extension member **128** and sensor slot **228** mating interface, together form the controlling index for proper x-axis and y-axis alignment of the interface-facing frame portion **116** (e.g., dual-feature x-y positioning scheme).

(98) However, in certain embodiment, the latching engagement of slider **170** with cassette recess **200** provides tight tolerances for z-axis positioning of the cassette body **110** and corresponding interface-facing frame portion **116** for operation with the cassette-facing surface **216** of cassette recess **200** (e.g., z-axis alignment or distances require for proper operation of the pressure sensors as well as valve actuators). It is to be understood that techniques of z-axis alignment or depth other than slider protrusion **174** positioning are contemplated. For example, positioning protrusion or pin may include a locking or stabilizing feature (e.g., a transverse aperture or engagement flanges) that may be removably coupled with a corresponding receiving port feature or mechanism in either the cassette **100** or cassette recess **200**. In other words, a snap engage type of component interfacing scheme.

(99) It is to be understood that positioning protrusion **220** may have various cross sectional geometries in addition to the circular cross-section protrusion/pin-type illustrated in FIG. 3C and other figures. For example, pin may have a triangular, square, or other polygonal cross-section. For instance, a polygonal shape may be used to provide tighter tolerances to the x-y positioning of the cassette body **110** when cassette **100** has an air-in-line features the does not protrude into and received by cassette recess **200**. Alternatively or in addition, the slider **170** may be configured have to tighter tolerances with respect to the x-axis and/or y-axis. For example, the x-axis tolerance may be substantially tight so as to avoid or substantially limit any minor or miniscule rotation of the cassette body **110** during prolonged continuous operation with infusion pump system **10**, **11**. Moreover, the positioning port **120** and positioning protrusion **220** mating interface can be reversed in some implementation (e.g., the positioning protrusion may be disposed on cassette **100**).

(100) However, in some implementations of cassette **100** and cassette recess **200**, a dual-feature x-y positioning scheme as described herein can be advantageous so as to avoid too much stress on a single mating feature. For example, positioning protrusion **220** having a circular cross-section cooperatively engaging with positioning port **120** can provide tight tolerances for either the x-axis or y-axis alignment of interface-facing frame portion **116**, but not necessarily both x-axis or y-axis alignment as rotation between positioning protrusion **220** and positioning port **120** can occur. However, in conjunction with a second x-y positioning such as fluid pathway extension member **128** and sensor slot **228** mating interface, tight tolerances can be achieved without undue stress on either of the x-y positioning features, as well as the cassette **100** and cassette recess **200** in general.

(101) Accordingly, advantages of the x-y positioning schemes disclosed herein include, but are not limited to, preventing sheering to sensor features (e.g., upstream pressure sensing probe **232** and downstream pressure sensing probe **234** may not engage with sensors until they are properly aligned by the x-y positioning features, aiding in full and proper valve operation (e.g., inlet-side valve actuator **222** and outlet-side valve actuator **224** may not fully close a corresponding valve if misaligned on the x-axis and/or y-axis occurs), and assisting with communicating a correct timing for latching slider **170** of cassette **100** with cassette recess **200** (e.g., caregiver can sense engagement of x-y positioning features between cassette **100** and cassette recess **200**).

(102) FIGS. 4A-4C illustrate an example of a cassette engagement sequence with cassette **100** and cassette recess **200**. Cassette **100** may be aligned such that the plurality of protrusions **174** on slider **170** may be aligned along z-axis **197** for engagement with the plurality of slots **274** of cassette recess **200**. In accordance with certain aspects, cassette **100** may have a longitudinal length along y-axis **195**, a lateral width along x-axis **196**, and a depth along z-axis **197**. As illustrated in FIG. 4A and described herein, the depth of the cassette **100** may be a smaller dimension than either the length or the width of cassette **100**. In this regard, cassette **100** is front loaded into cassette recess **200**.

(103) As illustrated in FIG. 4B, the plurality of protrusions **174** on slider **170** of cassette **100** may

engage with the plurality of slots **274** of cassette recess **200** such that cassette **200** is substantially seated (but not latched) within cassette recess **200**. Next, as illustrated in FIG. **4C**, slider **170** may be longitudinally articulated along y-axis **195** such that cassette **100** is latched within cassette recess **200**. It is to be noted that in some embodiments, loading of cassette **100** within cassette recess **200** may be laterally oriented (e.g., 90° rotation of z-axis **197** so that x-axis **196** and y-axis **195** are switched).

(104) With reference to the example cassette engagement sequence of FIGS. **4A-4C**, aspects of flow stop valve are provided with additional reference to the examples of FIGS. **5A** and **5B**. In certain embodiments, flow stop valve **164** may be configured to restrict and/or regulate fluid flow proximal to outlet **114** of cassette body **110**. In order to insert cassette **100** into cassette recess **200**, slider **170** should be articulated to a first position with respect to cassette body **110** (e.g., articulated upwardly in certain implementations; FIG. **4A**). In the first position, flow stop valve **164** is aligned under a portion (e.g., flat surface) of interface-facing slider section **176** of slider **170** (FIG. **5A**). When the slider **170** is positioned in the first position, the portion of interface-facing slider section **176** contacts and activates flow stop valve **164** such that fluid flow is occluded at that position of the fluid pathway proximal to outlet **114** cassette body **110**. Therefore, fluid leakage can be avoided during the final preparation stages (e.g., after priming of cassette **100**) and prior to insertion into cassette recess **200**. In accordance with certain configurations, the first position of slider **170** may correspond to a position of cassette **100** for engaging and disengaging with cassette recess **200**.

(105) During insertion of cassette **100**, once cassette body **110** is placed in cassette recess **200** (FIG. **4B**), slider **170** can be articulated to a second position with respect to cassette body **110** (e.g., articulated downwardly in certain implementations; FIG. **4C**). In this second position cassette **100** will be latched within cassette recess **200** by virtue of protrusions **174** being engaged with slots **274**. In the second position, flow stop valve **164** is aligned under a stop valve guard **178** (e.g., ramped or recessed surface) of interface-facing slider section **176** of slider **170** (FIG. **5B**). When the slider **170** is positioned in the second position, the portion of interface-facing slider section **176** does not contact flow stop valve **164** (or alternatively does not contact flow stop valve **164** sufficiently to activate flow stop valve **164**), and flow stop valve **164** operates to allow fluid to flow freely through flow stop valve **164** to outlet **114**.

(106) In accordance with certain embodiments, stop valve guard **178** may be positioned proximal to an edge along interface-facing slider section **176** of slider **170** such that when cassette **100** is securely latched or locked within cassette recess **200** (e.g., in the second position; FIG. **4C**), stop valve guard **178** is positioned above flow stop valve **164** (FIG. **5B**). In this regard, stop valve guard **178** can protect flow stop valve **164** from being inadvertently depressed and activated to restrict fluid flow while cassette **100** is in use. For example, a force applied to the slider side of cassette **100** while locked within cassette recess **200** would not depress flow stop valve **164** as the lateral tolerances of the slidable coupling between cassette **100** and slider **170** are tighter than a distance between the inner surface of the stop valve guard **178** and an outer surface of the flow stop valve **164**. In this regard, distances between stop valve guard **178** and flow stop valve **164** may be optimized such that anticipated forces applied to cassette **100** (e.g., from a user or caregiver inadvertently bumping cassette **100** or road vibrations in moving ambulance setting) may not cause an undesired activation of flow stop valve **164**.

(107) Similarly, when cassette **100** is to be disengaged from cassette recess **200**, slider **170** can be unlatched or unlocked from cassette recess **200** by accessing slider grip **172** and articulating slider **170** back to the first position with respect to cassette body **110** (FIG. **4B**). Some amount of force may be required by the user to articulate slider **170** to the first position as the plurality of protrusions **174** may be securely latched with corresponding slots **274** while the slider **170** is in the second position. Once slider **170** is in the first position, cassette **100** may be removed from cassette recess **200** by pulling slider grip **172** outwardly.

(108) Furthermore, slider **170** may be articulated from the first position to the second position

when cassette **100** is not engaged with cassette recess **200**. Thus, flow stop valve **164** will operate to allow fluid to flow freely through flow stop valve **164** to outlet **114**. Therefore, cassette **100** may be primed with fluid while being disengaged from cassette recess **200** when slider **170** is in the second position (but not latched with cassette recess **200**).

(109) It is to be understood that in other implementations, the stop valve guard feature can be a recess distal from an edge of interface-facing slider section **176**, or an aperture or slit extending through interface-facing slider section **176**, for example. Moreover, in some embodiments, flow stop valve and stop valve guard features may be positioned on the slider-facing side of cassette body **110**.

(110) The examples of FIGS. 5A-5D illustrate aspects of a flow stop valve. It is to be understood that various constructions of flow stop valve **164** are contemplated in the present disclosure. For example, in accordance with certain embodiments, striker **181** is configured to redirect the longitudinally introduced force applied by movement interface-facing slider section **176** into a more generally orthogonal force applied to top membrane portion **183** of flow stop valve **164**. In some aspects, striker **181** can be a hard plastic (e.g., polycarbonate) dome-shaped piece disposed over and co-molded with top membrane portion **183**. In other aspects, striker **181** may be constrained in a manner for engagement with top membrane portion **183** (e.g., situated in an aperture above top membrane portion **183**). As such, potential sheering forces longitudinally applied directly to the membrane portion **183** of the flow stop valve **164** may be avoided in accordance with certain embodiments. Flow stop valve **164** may include through-hole base portion **185** (e.g., through base portion **119**) for releasable engagement with top membrane portion **183**. Top membrane portion **183** urged by striker **181** is configured to circumferentially seal through-hole base portion **185** so that fluid flow is restricted to outlet **114** in accordance with certain aspects.

(111) In accordance with some embodiments, slider **170** may interface with flow stop valve **164** to regulate the fluid flow therethrough. As illustrated in the examples of FIGS. 5E and 5F, interface-facing slider section **176** may include one or more cassette-facing detents **179** for contacting striker **181** such that top membrane portion **183** partially occludes through-hole base portion **185** to regulate fluid flow to outlet **114**. The one or more cassette-facing detents **179** may be longitudinally aligned corresponding to slider **170** motion with respect to striker **181**. Accordingly, the resulting reduced fluid flow through outlet **114** may be beneficial during priming and preparation procedures when cassette **100** is not engaged with cassette recess **200**. In some embodiments, when striker **181** makes contact with the one or more cassette-facing detents **179**, an audible click may be heard or discernable vibration may be felt by the user or caregiver during the slider engagement process.

(112) It is to be appreciated that in addition to the other noted advantages, the flow stop valve features described herein provide a flow stop operation of cassette **100** that does not involve interactions with infusion pump system **10**, **11**, cassette recess **200** or any mechanical or electrical component thereof.

(113) FIGS. 6A and 6B illustrate another example of a disposable IV cassette **300** and corresponding cassette recess **400** of an interface module. In accordance with other embodiments, cassette **300** may comprise a cassette body **310** and a slider **370**. Cassette **300** may be operatively coupled to cassette recess **400**.

(114) In accordance with some embodiments, cassette **300** may comprise a cassette body **310** and a slider **370**. Slider **370** can be fixably and slidably engaged with cassette body **310** such that slider **370** may articulate longitudinally with respect to cassette body **310**, but will be constrained within range of sliding motion such that the slider remains coupled to the cassette body **310**. In some embodiments, cassette **300** may be configured so that slider **370** does not extend around cassette body **310** (FIG. 6B).

(115) In some embodiments, slider **370** includes a slider grip **372** and a plurality of protrusion **374** that are configured to be releasably lockable with a plurality of slots **474** of the cassette recess **400**

(e.g., L-shaped locking channels). The plurality of protrusions **374** may be disposed at various locations on slider **370**.

(116) Edges **352** of cassette **300** and corresponding perimeter of cassette recess **400** may include at least one arcuate edge **352a** contrasted with at least one opposite corner edge **352b**, for example. Therefore, orientation of cassette **300** with respect to loading engagement with corresponding arcuate perimeter edge **452a** and corner edge **452b** of cassette recess **400** may be readily apparent to a user or caregiver.

(117) In accordance with some aspects, an overall size of cassette **300** and cassette recess **400** may be reduced. For example, in some embodiments, cassette body **310** may extended longitudinally a length between 65 mm and 75 mm, for example, by utilizing a single pump chamber/reservoir for sensing probe. Additionally, cassette body **310** may extended laterally a width between 34 mm and 39 mm, and may extend a depth between 10 mm and 14 mm. Fluid pathway extension member **328** may further extend between 8 mm to 10 mm. In some aspects, slider grip **372** may extend between 10 mm to 14 mm from cassette body **310**.

(118) For orientation reference with respect to the examples illustrated of FIG. **6A**, longitudinal axis or y-axis **395**, latitudinal axis or x-axis **396**, and depth axis or z-axis **397** are provided. In this regard, depth aspects of cassette **300** is shown in the example of FIG. **6A**.

(119) The process of cleaning of inlet recess **412**, outlet recess **414**, and cassette recess **400** is made efficient by the shallow recess configuration in accordance with some embodiments should any fluid or debris accumulate within cassette recess **400**. For example, the plurality of slots **474** arranged within cassette recess **400** can be reached and cleaned with a swab.

(120) The shallow recess configuration of cassette recess **400**, and associated longitudinal alignment of cassette **300** such that a smaller of volumetric dimensions of cassette **300** (e.g., depth being smaller than length and width in some embodiments) further enables additional space for arrangement of mechanical couplings and operational interfaces and optimizes the overall space requirements of cassette recess **400** and infusion pump system, or the like, in general.

(121) Various types, placement, and orientations of the plurality of protrusions **374** disposed on slider **370** are contemplated as described herein. Moreover, in accordance with some aspects, features of cassette recess **400** are designed to avoid wear down and/or risk of malfunction. For example, the plurality of slots **474** arranged within cassette recess **400** may not include any movable latching mechanism in some embodiments as such movable latching mechanisms may be susceptible to excessive wear and mechanical failure over repeated use with multiple disposable IV cassettes **300**.

(122) Cassette **300** can be loaded into cassette recess **400**. In this regard, the loading of the interface-facing side of cassette body **310** can avoid shear forces applied to the sensors, alignment features, and other engaging interfaces of cassette body and cassette-facing surface **416** of cassette recess **400**.

(123) With reference to the examples of FIG. **6B**, cassette body **310** may comprise interface-facing frame portion **316** and slider-facing base portion **319** with membrane **317** disposed substantially therebetween (e.g., portions of membrane **317** may extend through some openings of frame portion **316**). In accordance with some embodiments, membrane **317** can be a compliant material co-molded to the base portion **319** and sealingly engaged with frame portion **316** for defining a fluid pathway through cassette body **310** from inlet **312** to outlet **314**. In some embodiments, membrane **317** may also be co-molded to striker **381** for defining, in part, a flow stop valve **364**.

(124) Mating edges of frame portion **316** and base portion **319** may be connected by fusing, welding, gluing, or the like. Membrane **317** and base portion **319** may further define a plurality of other features, some of which may be accessed through openings in frame portion **319**.

(125) Frame portion **316**, membrane **317**, and/or base portion **319** may define features in or along the fluid pathway, in accordance with some embodiments.

(126) For example, beginning from inlet **312**, the fluid pathway may include features such as, but

not limited to, inlet-side valve **322**, pump chamber/sensing reservoir **326**, outlet-side valve **324**, fluid pathway extension member **328**, and flow stop valve **364**. Other features that are not in or along the fluid pathway, but are disposed on cassette body **310**, may include positioning port **320**. With respect to extension member **328**, a portion of the fluid pathway can be extended away or protrude orthogonally from the generally flat and planar exterior surface of interface-facing frame portion **316** so as to make the fluid in the fluid pathway available for other detection techniques performed by infusion pump system **10, 11**.

(127) As illustrated, fluid pathway extension member **328** may be formed from orthogonally extending portions of frame portion **316**, membrane **317**, and base portion **319**. However, in other embodiments, a fluid pathway access point may be configured for air-in-line detection, for example, from a section of the exterior surface of interface-facing frame portion **316**.

(128) One or more fluid sensors may be disposed within sensor slot **428** such as ultrasonic sensors configured as an air-in-line detector. In some embodiments, extension member **328** may be disposed on cassette body **310** and positioned along the fluid pathway between pump chamber/sensing reservoir **326** and flow stop valve **364**.

(129) Cassette body **310** may include a pump drive assembly in accordance with some embodiments. For example, the pump drive assembly may include pump drive mechanism **342** for receiving pump actuator **442** of cassette recess **400**. Pump drive mechanism **342** can be operatively coupled to piston **345** slidably engaged within piston guide **343** casing (e.g., generally cylindrical or frustoconical casing) such that reciprocal movement of piston **345** may change a total volume of pump chamber pump chamber/sensing reservoir **326** thereby urging fluid through the fluid pathway of cassette body **310**.

(130) In some embodiments, pump drive assembly may be configured to produce a 3.5 mm piston stroke for operation with pump chamber/sensing reservoir **326** configured to be a 10 mm outer diameter reservoir. Moreover, pump drive assembly may be arranged below pump chamber/sensing reservoir **326**, in accordance with some embodiments.

(131) With cassette **300** primed and secured in cassette recess **400**, an example pumping operation may comprise activating outlet-side valve actuator **424** such that outlet-side valve **324** is closed or sealed while activating inlet-side valve actuator **422** such that inlet-side valve **322** is opened.

Opening of inlet-side valve **322** may coincide with or occur shortly after a reverse stroke of piston **345** (e.g., a movement of piston **345** away from pump chamber/sensing reservoir **326**).

Accordingly, fluid can flow from inlet **312** to pump chamber/sensing reservoir **326**.

(132) Pumping operation may further comprise activating outlet-side valve actuator **424** such that outlet-side valve **324** is open while activating inlet-side valve actuator **422** such that inlet-side valve **322** is closed or sealed. Opening of outlet-side valve **324** may coincide with or occur shortly before a forward stroke of piston **345** (e.g., a movement of piston **345** toward pump chamber/sensing reservoir **326** such that contact is by a head of piston **345** is made with a portion of pump chamber/sensing reservoir **326**). Thus, fluid can flow from pump chamber/sensing reservoir **326** to outlet **314**.

(133) In some embodiments, cassette recess **400** may include pump chamber sensing probe **426** to enable measurement of in-line pressure. For example, pump chamber sensing probe **426** may operably contact pump chamber/sensing reservoir **326** sized to be sufficient for pressure sensing through a corresponding opening of interface-facing frame portion **316**.

(134) As illustrated in the example of FIG. 6A, cassette **300** may be aligned such that the plurality of protrusions **374** on slider **370** may be aligned along z-axis **397** for engagement with the plurality of slots **474** of cassette recess **400**. In accordance with some aspects, cassette **300** may have a longitudinal length along y-axis **395**, a lateral width along x-axis **396**, and a depth along z-axis **397**. The depth of cassette **300** may be a smaller dimension than either the length or the width of cassette **300**. In this regard, cassette **300** is loaded into cassette recess **400**.

(135) In accordance with some examples, the x-y positioning of cassette **300** within cassette recess

400 can be constrained by the positioning port **320** and positioning protrusion **420** mating interface, as well as fluid pathway extension member **328** and sensor slot **428** mating interface. In this regard, cassette **300** and cassette recess **400** can have two point of contact in the z-axis direction (e.g., an axis through and transverse to a general plane of the interface-facing surface of cassette body **310** of cassette **300** and cassette-facing surface **416** of cassette recess **400**) for interlock alignment of the cassette with respect to the x-y positioning of the interface side of the cassette body **310**.

(136) In some embodiments, positioning port **320** may be located proximal to inlet-side valve **322** and aligned above pump chamber/sensing reservoir **326**. Positioning protrusion **420** may be located proximal to inlet-side valve actuator **422** and aligned above pump chamber sensing probe **426**.

Accordingly, the cassette **300** may be positioned properly to aid in sensing accuracy for a single sensing reservoir implementation without over constraining the cassette **300**.

(137) In some embodiments, the positioning port **320** and positioning protrusion **420** mating interface, and fluid pathway extension member **328** and sensor slot **428** mating interface, together form the controlling index for proper x-axis and y-axis alignment of the interface-facing frame portion **316** (e.g., dual-feature x-y positioning scheme).

(138) Flow stop valve **364** may be configured to restrict and/or regulate fluid flow proximal to outlet **314** of cassette body **310**, in accordance with some embodiments. In order to insert cassette **300** into cassette recess **400**, slider **370** should be articulated to a first position with respect to cassette body **310** (e.g., articulated upwardly in some implementations). In the first position, flow stop valve **364** may be aligned under a portion (e.g., flat surface) of interface-facing slider section **376** of slider **370**. When the slider **370** is positioned in the first position, the portion of interface-facing slider section **376** contacts and activates flow stop valve **364** such that fluid flow is occluded at that position of the fluid pathway proximal to outlet **314** cassette body **310**.

(139) Therefore, fluid leakage can be avoided during the final preparation stages (e.g., after priming of cassette **300**) and prior to insertion into cassette recess **400**. During insertion of cassette **300**, once cassette body **310** is placed in cassette recess **400**, slider **370** can be slid to a second position with respect to cassette body **310** (e.g., articulated downwardly in some implementations).

(140) In the second position, flow stop valve **364** may be aligned under a stop valve guard **378** (e.g., ramped or recessed surface) of interface-facing slider section **376** of slider **370**. When the slider **370** is positioned in the second position, the portion of interface-facing slider section **376** does not contact flow stop valve **364** (or alternatively does not contact flow stop valve **364** sufficiently to activate flow stop valve **364**), and flow stop valve **364** operates to allow fluid to flow freely through flow stop valve **364** to outlet **314**. Therefore, cassette **300** may be operable for fluid flow rate regulation via cassette recess **400**.

(141) In accordance with some embodiments, stop valve guard **378** may be positioned proximal to an edge along interface-facing slider section **376** of slider **370** such that when cassette **300** is securely latched or locked into cassette recess **400** (e.g., in the second position), stop valve guard **378** is positioned above flow stop valve **364**. In this regard, stop valve guard **378** can protect flow stop valve **364** from being inadvertently depressed and activated to restrict fluid flow while cassette **300** is in use. For example, a force applied to the slider side of cassette **300** while locked within cassette recess **400** would not depress flow stop valve **364** as the lateral tolerances of the slidable coupling between cassette **300** and slider **370** may be tighter than a distance between the inner surface of the stop valve guard **378** and an outer surface of the flow stop valve **364**. In this regard, distances between stop valve guard **378** and flow stop valve **364** may be optimized such that anticipated forces applied to cassette **300** and/or cassette recess **400** may not cause an undesired activation of flow stop valve **364**.

(142) Similarly, when cassette **300** is to be disengaged from cassette recess **400**, slider **370** can be unlatched or unlocked from cassette recess **400** by accessing grip **372** and articulation slider **370** back to the first position with respect to cassette body **310**. Some amount of force may be required by the user or caregiver to move slider **370** to the first position as the plurality of protrusions **374**

may be securely latched with corresponding slots **474** while the slider **270** is in the second position. Once slider **370** is in the first position, cassette **300** may be removed from cassette recess **400** by pulling grip **372** outwardly.

(143) Slider **370** may be articulated from the first position to the second position when cassette **300** is not engaged with cassette recess **400** or interface module. Thus, flow stop valve **364** will operate to allow fluid to flow freely through flow stop valve **364** to outlet **314**. Therefore, cassette **300** may be primed with fluid while being disengaged from cassette recess **400** when slider **370** is in the second position (but not latched within cassette recess **400**).

(144) In accordance with some embodiments, striker **381** of flow stop valve **364** is configured to redirect the longitudinally introduced force applied by movement interface-facing slider section **376** into a more generally orthogonal force applied a top membrane portion or other valve actuation mechanism of flow stop valve **364**. In some aspects, striker **381** can be a hard plastic (e.g., polycarbonate) dome-shaped piece disposed over and co-molded the top membrane portion.

(145) In accordance with some embodiments, slider **370** may interface with flow stop valve **364** to regulate the fluid flow therethrough. Interface-facing slider section **376** may include one or more cassette-facing detents or a ramp or incline portion fixably movable with respect to striker **381** such that flow stop valve **364** can be partially occluded to regulate fluid flow to outlet **314**.

(146) It is to be further appreciated that cassette **100**, **300** and the various features thereof may be constructed from a few number of components (e.g., formed from four to seven parts in accordance with certain embodiments) as described herein given the benefit of the present disclosure.

(147) FIGS. 7A and 7B illustrate examples of a disposable IV pump cassette **500** and corresponding cassette recess **600** of an interface module. In accordance with certain embodiments, cassette **500** may comprise a cassette body **510** and a slider **570**. Cassette **500** may include certain visual indicators related to operation aspects of the cassette and the infusion pump system in general. For example, cassette may include identifiable images such as arrows indicating that the slider **570** may articulate longitudinally **591** with respect to cassette body **510** and a patient figure proximal to outlet **514**. In accordance with some aspects, cassette **500** may include one or more lens areas **573** for magnification of the fluid pathway within the cassette body **510**. The one or more lens areas **573** may be disposed on the slider **570** extending at least a portion of the fluid pathway within the cassette body **510**. In some embodiments, the one or more lens areas **573** may extend longitudinally along the slider **570**. In accordance with some aspects, the one or more lens areas **573** may provide magnification for greater than approximately 50% of the fluid pathway extending within the cassette body **510** from inlet **512** to outlet **514**. In this regard, during priming or prepping a cassette **500**, as well as during operation of the cassette **500**, a user or caregiver may use the one or more lens areas **573** to ensure that any visible air bubbles throughout a substantial portion or the fluid pathway have been removed and fluid is flowing properly. In accordance with some aspects, one or more cassette-seated sensors may be disposed within the cassette recess **600** so as to inform central processing unit **12** that the cassette is locked or secured into place within the cassette recess **600** or seat.

(148) Slider **570** can be fixably and slidably engaged with cassette body **510** such that slider **570** may articulate longitudinally **591** with respect to cassette body **510**, but may be constrained within range of sliding motion such that the slider remains coupled to the cassette body **510**, for example. Slider **570** may be formed from rigid plastic or polymer material having lubricating characteristics (e.g., incorporating silicon or polytetrafluoroethylene (PTFE) additives), and is clear or translucent in accordance with certain embodiment. In some embodiments, slider **570** may be polycarbonate. Slider **570** includes a slider grip **572** or handle portion and a plurality of protrusion **574** or lugs that are configured to be releasably lockable with a plurality of slots **674** of the cassette recess **600** (e.g., L-shaped locking channels). In this regard, cassette **500** can be self-latched into the cassette recess **600**. Accordingly, a door or lever action is not required in order to retain the cassette **500** within the cassette recess **600**. As such, when the cassette is used in an infusion pump system in accordance

with certain aspects of the present disclosure, a front area of the cassette **500** including grip **572** can be accessed during operation of the infusion pump system.

(149) Similar to other described embodiments, the plurality of protrusions **574** may be disposed at various locations on slider **570**. As illustrated in the example of FIG. 7C, slider **570** may comprise a first pair of protrusions **574a** and a second pair of protrusions **574b**. The first pair of protrusions **574a** may be spaced apart a similar distance (e.g., width **W3**) as the second pair of protrusions **574b**. Thus, opposite sides (e.g., left and right sides according to an example orientation) of cassette **500** can be substantially parallel in certain embodiments. In accordance with some aspects, **W3** may be between 21 mm and 25 mm in some embodiments. Cassette **500** may comprise a first end portion **553a** (e.g., top portion according to an example orientation) of cassette body **510** that is shaped differently than a second end portion **553b** (e.g., bottom portion according to an example orientation) of cassette body **510**.

(150) For example, first end **553a** can be slightly rounded or squared off with respect to the sides of cassette body **510**, whereas second end **553b** can be angled or otherwise generally pointed (e.g., indicative of a direction of fluid flow through cassette **500**). Inlet **512** may be substantially longitudinally aligned with outlet **514**, in accordance with certain embodiments. In this regard, cassette **500** orientation within cassette recess **600** is clear to a user or caregiver such that a cassette **500** is not inadvertently installed (or inadvertently positioned when being primed) in an inverted manner.

(151) Accordingly, perimeter edges **552** of cassette **500** and corresponding perimeter edges of cassette recess **600** may include one or more perimeter edges for engagement between cassette **500** and cassette recess **600**. For example, the perimeter edges **552** may correspond to a perimeter opening of cassette recess **600** for front-loading cassette **500** into cassette recess **600**. In accordance with certain implementations, front-loading a cassette means that the cassette engages with a cassette recess disposed on a surface of an interface unit in a direction of the smallest of the cassette's three general dimensions (e.g., a depth along a z-axis as illustrated in the figures). In this regard, a surface of an interface unit may be on located on a front, side, or rear surface with respect to a general front of the interface unit (e.g., where the electronics and display are located) and still be considered to be front-loading surface. Additionally, in some implementation, a front-loaded cassette may be readily accessible to a user or caregiver to remove such that no cover prohibits access to the installed cassette. However, in other implementations, a locked or latched cover may exist on the interface unit to protect or limit access to the installed cassette. It is to be further understood that the front-loaded cassette need not be installed with the cassette's longitudinal axis extending vertically with respect to an upright position of the interface unit. For example, some interface units may include cassette recesses oriented such that the longitudinal axis of the front-loading cassette is oriented horizontally with respect to the interface unit.

(152) With reference to FIGS. 7C and 7D, cassette body **510** (with or without slider **570**) may comprise top lateral edge **552a**, straight longitudinal side edge **552b**, angled bottom edge **552c**, angled bottom edge **552d**, and straight longitudinal side edge **552e**. Straight longitudinal side edge **552b** and straight longitudinal side edge **552e** may be generally parallel and opposing edges as described herein, in accordance with certain example aspects.

(153) First and second end portion **553a**, **553b**, may provide benefit by enabling visual identification as well as prohibiting improper orientation or seating of cassette **500** with a corresponding cassette recess **600**. Additionally, minimizing a size of cassette **500** and optimizing a surface area of one or more corresponding cassette recess **600** may be achieved by the elongate longitudinal design of cassette **500**. Moreover, in certain embodiments, inlet **512** may be substantially longitudinally aligned with outlet **514** with the fluid pathway extending in a generally straight line from inlet **512** to outlet **514**, thereby simplifying user or caregiver troubleshooting of cassette **500** during operation of infusion pump system **10**, **11**, as well as allowing free flow of fluid during priming of cassette **500**.

(154) In some embodiments, cassette **500** may include detent **511** and corresponding recess **571**. For example, detent **511** may be positioned on or proximate to inlet **512** be positioned on the slider (155) An overall size of cassette **500** and cassette recess **600** may be reduced, in accordance with some aspects. For example, in certain embodiments, cassette body **510** may extend longitudinally a length (L3) between approximately 85 mm and 128 mm. In this regard, cassette body **510** may be dimensioned as having an elongate body section with a length larger than a width. For orientation reference with respect to the various views of the examples illustrated of FIGS. 7C and 7D, longitudinal axis or y-axis **195**, latitudinal axis or x-axis **196**, and depth axis or z-axis **197** are provided.

(156) In this regard, depth aspects of cassette **500** is shown in the example of FIG. 7D. For example, in certain embodiments, cassette body **510**, or a substantial portion thereof, may extend depth (D3) between 6 mm and 8 mm. Fluid pathway extension member **528** may further extend between 8 mm to 10 mm, and pump structure **541** may further extend between 10 mm to 16 mm, in accordance with some embodiments. In certain aspects, slider grip **572** may extend between 10 mm to 14 mm from cassette body **510**. It is to be appreciated that the process of cleaning of inlet recess **612**, outlet recess **614**, and cassette recess **600** is made efficient in the shallow recess configuration in accordance with certain embodiments should any fluid or debris accumulate within cassette recess **600**. The shallow recess configuration of cassette recess **600**, and associated longitudinal alignment of cassette **500** such that a smaller of volumetric dimensions of cassette **500** (e.g., depth being smaller than length and width in certain embodiments) further enables additional space for arrangement of mechanical couplings and operational interfaces and optimizes the overall space requirements of cassette recess **600** and infusion pump system in general.

(157) Various types, placement, and orientations of the plurality of protrusions **574** disposed on slider **570** are contemplated in the present disclosure. In accordance with certain aspects, features of cassette recess **600** are designed to avoid wear down and/or risk of malfunction. For example, the plurality of slots **674** arranged within cassette recess **600** may be devoid of any movable latching mechanism in certain embodiments as such movable latching mechanisms may be susceptible to excessive wear and mechanical failure over repeated use with multiple disposable IV cassettes **500**. In operation, cassette **500** can be loaded directly into cassette recess **600**. In this regard, the direct loading of the cassette **500** will enable avoidance of shear forces that might otherwise be applied to the sensors, alignment features, and other engaging interfaces of cassette-facing surface **616** of cassette recess **600** from interaction with the interface-facing side of cassette body **510** as it is loaded into cassette recess **600**.

(158) It is to be understood that modification to the various features of cassette **500** can be made to accommodate the various cassette features and coupling techniques of other embodiments disclosed herein.

(159) Referring now to the examples of FIGS. 8A-8C, cassette body **510** may comprise interface-facing frame portion **516** and slider-facing base portion **519** with membrane **517** disposed substantially therebetween (e.g., portions of membrane **517** may extend through some openings of frame portion **516**). In accordance with certain embodiments, membrane **517** can be a compliant material co-molded to the frame portion **516** and sealingly engaged with base portion **519** for defining a fluid pathway through cassette body **510** from inlet **512** to outlet **514**. Mating edges of frame portion **516** and base portion **519** may be connected by fusing, welding, gluing, or the like. Membrane **517** and base portion **519** may further define a plurality of other features, some of which may be accessed through openings in frame portion **516**.

(160) Frame portion **516**, membrane **517**, and/or base portion **519** may define features in or along the fluid pathway, in accordance with certain embodiments. For example, beginning from inlet **512**, the fluid pathway may include features such as, but not limited to, upstream pressure dome **532** (e.g., an inlet-side compliant reservoir), inlet-side valve **522**, pump chamber **525** (e.g., within a pump chamber wall **543** of pump structure **541**), outlet-side valve **524**, a downstream pressure

dome **534**, (e.g., an outlet-side compliant reservoir), fluid pathway extension member **528**, and flow stop valve **564**.

(161) In certain embodiments, for example, first pair of protrusions **574a** and second pair of protrusion **574b** may be disposed on slider **570** such that when slider **570** is engaged with and locked into cassette recess **600** and corresponding slots **674**, the first pair of protrusions **574a** is positioned or laterally aligned with upstream pressure dome **532** and corresponding inlet-side pressure sensing probe recess **632a**, and the second pair of protrusions **574b** is positioned or laterally aligned with downstream pressure dome **534** and corresponding outlet-side sensing probe recess **634a**. In this regard, structural integrity of cassette **500** and pressure sensing operations may be optimized in accordance with aspects of the present disclosure.

(162) Other features of cassette **500** disposed on cassette body **510**, may include pump structure **541** having an exterior support wall **574** that can be used as a positioning protrusion, for example, and slider stopper **551**. With respect to extension member **528**, a portion of the fluid pathway can be extended away or protrude orthogonally from the generally flat and planar exterior surface of interface-facing frame portion **516** so as to make the fluid in the fluid pathway available for certain detection techniques performed by infusion pump system **10**, **11**. As illustrated in the example of FIGS. **8A-8C**, fluid pathway extension member **528**, as well as pump structure **541**, may be formed from orthogonally extending portions of frame portion **516**, membrane **517**, and/or base portion **519**.

(163) In accordance with certain embodiments, membrane **517** may be formed from a thermoplastic elastomer (TPE). Characteristics of certain TPEs can enable effective co-molding with other materials, for example, polycarbonate. Accordingly, in some embodiments, membrane **517** may be co-molded to frame portion **516** and striker **581** may be co-molded to a portion of membrane **517** defining a flow stop valve **564**. However, in some embodiments, membrane **517** can be formed from silicon, a silicon-based compound, an elastomeric material suitably compliant for fluid flow, or the like.

(164) In accordance with certain embodiments, interface-facing frame portion **516** and slider-facing base portion **519** may be formed from a rigid plastic such as, but not limited, a polycarbonate. Additionally, the rigid plastic of frame portion **516** and base portion **519** may be clear or translucent. The material of membrane **517** (e.g., TPE or other compliant material) and rigid plastic slider **570** may also be clear or translucent, thereby allowing a user or caregiver to readily observe fluid passage through a substantial portion of the fluid pathway of cassette body **510**. In some embodiments, the fluid pathway portion of cassette body **510** will be clear or translucent, and other portions will be frosted so as to direct a user or caregiver's attention to the fluid pathway.

(165) In some implementations, slider **570**, base portion **519**, and membrane **517** may be clear or translucent (or at least some portions along the fluid pathway), and the frame portion **516** may not be translucent. For example, the frame portion **516** may be colored in a manner so as to contrast against a color or tint of the fluid expected to be used with cassette **500**. In some embodiments, one or more lens areas **573** may be disposed on base portion **519** alternatively, or in addition to, one or more lens areas **573** disposed on slider **570**.

(166) With additional reference to the example of FIG. **8D**, one or more fluid sensors may be disposed within sensor slot **628**. The one or more fluid sensors disposed within sensor slot **628** can be ultrasonic sensors configured with an air-in-line detector, for example. In certain embodiments, extension member **528** may be disposed on cassette body **510** and positioned along the fluid pathway between downstream pressure dome **534** and flow stop valve **564**. However, in some embodiments, extension member **528** can be positioned at other locations along the fluid pathway. Additionally, in other embodiments, a plurality of extension members **528** with a plurality of corresponding sensor slots **628** may be positioned along a fluid pathway of cassette body **510**.

(167) As illustrated in the examples of FIG. **8A-8E**, cassette body **510** may include a pump structure **451** in accordance with certain embodiments. For example, the pump structure **541** may

include piston assembly **542** for receiving pump actuator **642** of cassette recess **600**. As such, cassette **500** with pump structure **541** may be configured for orthogonal piston assembly operation whereby a pump actuator of an interface module applies a force orthogonal to a general plane of interface-facing frame portion **516**. Piston assembly **542** can include piston **545** co-molded with elastomeric seal member **546**; however, other piston assemblies are contemplated under the present disclosure. Elastomeric seal member **546** may be secured to a portion of support wall **547** (e.g., an outer rim portion) distal to a general plane of the interface-facing surface of cassette body **510** so that piston assembly **542** may be slidably engaged with pump chamber wall **543**. A chamber seal surface **546a** of elastomeric seal member **546** that is internal to the pump structure **541** may be slidably engaged with an internal surface of the pump chamber wall **543** such that the seal surface **546a** constitutes a moveable wall portion of pump chamber **525**. Chamber base section **519a** of base portion **519** may constitute an opposing stationary wall portion of the pump chamber **525**. (168) In this regard, pump chamber **525** may be defined by pump chamber wall **543**, chamber base section **519a**, and seal surface **546a**, in accordance with certain embodiments. Thus, movement of the piston assembly **542** and associated seal surface **546a** can change a volume of pump chamber **525** to urge fluid through the fluid pathway of cassette body **510**. As such, the fluid pathway may be controllable by operation of the pump chamber **525**, inlet-side valve **522**, outlet-side valve **524**, and flow stop valve **564**, for example.

(169) In certain embodiments, slider grip **572** (or like portion of slider **570**) may be positioned such that slider grip **572** is proximate to pump structure **541** (e.g., laterally respect to z-axis **197**) such that a single hand of a user or caregiver can be used to prime cassette and articulate the slider **570** to control fluid flow. For example, a user or caregiver may place and secure slider grip **572** between two fingers of a first hand and use the thumb of the first hand to manually pump the pump structure **541** by pressing on an exterior portion of the piston assembly **542**. The user or caregiver may apply a force to the exterior portion of the piston assembly **541** that is substantially orthogonal to a general plane of an interface-facing surface of the pump cassette **500**, for example. The user or caregiver may articulate slider **570** with leverage from the thumb of the first hand in contact with the pump structure **541** (or proximate thereto) from a first position to a second position to occlude or allow fluid flow through cassette **500** when the cassette **500** is not engaged with cassette recess **600**. Thus, in some embodiments, slider grip **572** may be positioned on slider **570** such that slider grip **572** is proximal to pump structure **541** and distal from inlet **512** in any articulable position of the slider **570** (e.g., from the first position to the second position) operably engaged with cassette body **510**. In this regard, the pump cassette **500** may be ergonomically designed for quick priming.

(170) Pumping operation of infusion pump system **10**, **11** when cassette **500** is primed and seated in cassette recess **600** may comprise activating outlet-side valve actuator **624** (not shown in FIG. **8C**) in outlet-side valve actuator slot **624a** such that outlet-side valve **524** is closed or sealed while activating inlet-side valve actuator **622** (not shown in FIG. **8C**) in inlet-side valve actuator slot **622a** such that inlet-side valve **522** is opened. Opening of inlet-side valve **522** may coincide with or occur shortly after a reverse stroke of piston **545** of piston assembly **542** (e.g., a movement of piston **545** and associated seal surface **546a** away from chamber base section **519a**, thereby enlarging a volume of pump chamber **525**). Accordingly, fluid can flow from upstream pressure dome **532** to pump chamber **525**. Alternatively, outlet-side valve **524** may comprise a one-way valve mechanism that permits flow of fluid under normal conditions in one direction (from a fluid container to a patient). Additionally, inlet-side valve **522** may also comprise a one-way valve mechanism permitting flow of fluid in one direction (from a fluid container to a patient) under normal operating conditions. In this configuration, cassette recess **600** would not need to incorporate either outlet-side valve actuator **624** or inlet-side valve actuator **622**. Outlet-side valve **524** and inlet-side valve **522** may limit flow of fluid in one direction, but permit flow in an opposite direction in the event fluid pressure overcomes a cracking pressure of the valves.

(171) Continuing with the valve-operated implementation, pumping operation may comprise

activating outlet-side valve actuator **624** such that outlet-side valve **524** is open while activating inlet-side valve actuator **622** such that inlet-side valve **522** is closed or sealed. Opening of outlet-side valve **524** may coincide with or occur shortly before a forward stroke of piston **545** (e.g., a movement of piston **545** and associated seal surface **546a** toward chamber base section **519a**, thereby reducing a volume of pump chamber **525**). Thus, fluid can flow from pump chamber **525** to downstream pressure dome **534** and consequently urging fluid out outlet **514**.

(172) In certain embodiments, compliant upstream pressure dome **532** and/or downstream pressure dome **534** can address any back pressure issues in the IV set, thereby allowing for an accurate and precise volume of fluid entering pump chamber **525** to be pumped. Additionally, in some embodiments, downstream pressure dome **534** may be larger (e.g., 12 mm) than upstream pressure dome **532** (e.g., 8 mm) so as to address more frequent pressure changes (e.g., temporary or minor occlusions) on the patient side of the IV set, for example.

(173) As illustrated in the example of FIGS. **8B** and **8E**, inlet-side valve **522**, pump chamber **525**, and outlet-side valve **524** may be longitudinally aligned within cassette body **510**, in accordance with certain embodiments. Thus, width and valve operation aspects may be further optimized. It is to be appreciated that orthogonal piston pump techniques as described with respect to cassette **500** and cassette recess **600** can provide repeatedly precise positive displacement of fluid in the pump chamber **525**.

(174) FIG. **8F** is a cross-sectional view of illustrating aspect of flow stop valve **564**. In accordance with certain embodiments, striker **581** is configured to redirect the longitudinally introduced force applied by movement interface-facing slider section **576** into a more generally orthogonal force applied to top membrane portion **583** of flow stop valve **564**. In some aspects, striker **581** can be a hard plastic (e.g., polycarbonate) dome-shaped piece disposed over and co-molded with top membrane portion **583**. In other aspects, striker **581** may be constrained in a manner for engagement with top membrane portion **583** (e.g., situated in an aperture above top membrane portion **583**). As such, potential sheering forces longitudinally applied directly to the membrane portion **583** of the flow stop valve **564** may be avoided in accordance with certain embodiments. Flow stop valve **564** may include through-hole base portion **585** (e.g., through base portion **519**) for releasable engagement with top membrane portion **583**. Top membrane portion **583** urged by striker **581** is configured to circumferentially seal through-hole base portion **585** so that fluid flow is restricted to outlet **514** in accordance with certain aspects.

(175) In accordance with some embodiments, a portion of slider **570** may interface with flow stop valve **564** to regulate the fluid flow therethrough. As illustrated in the example of FIG. **8F**, the portion of interface-facing slider section **576** does not contact flow stop valve **564** (or in some implementations does not contact flow stop valve **564** sufficiently to activate flow stop valve **564**), and flow stop valve **564** operates to allow fluid to flow freely through flow stop valve **564** to outlet **514**.

(176) In accordance with certain embodiments, stop valve guard **578** may be positioned proximal to an edge along interface-facing slider section **576** of slider **570** such that when cassette **100** is securely latched or locked within cassette recess **600**, stop valve guard **578** is positioned above flow stop valve **564**. In this regard, stop valve guard **578** can protect flow stop valve **564** from being inadvertently depressed and activated to restrict fluid flow while cassette **500** is in use.

(177) As illustrated in FIGS. **9A** and **9B**, slider **570** may be longitudinally articulated such that cassette **500** is latched within cassette recess **600** in accordance with certain orientations of the cassette **500** and cassette recess **600**. Cassette recess **600** may include interface structures operable in conjunction with infusion pump systems **10**, **11**. For example, pump actuator **642** may be configured to provide a force aligned with piston **545** that is substantially orthogonal to a general plane of the interface-facing surface of the pump cassette **500** in certain implementations. In such implementations, pump actuator **642** may engaged with the elastomeric seal member **546** when pump structure **541** is positioned in pump structure recess **641a** so that the piston **545** is retracted

when the pump actuator **642** performs a reverse stroke (e.g., away from the cassette **500**). However, in some implementations, the elastomeric seal member **546** provides sufficient resiliency or counter force to retract piston **545**.

(178) In addition to pump structure recess **641a**, cassette recess **600** includes inlet-side valve actuator slot **622a** and outlet-side valve actuator slot **624a**, which are operatively coupled to inlet-side valve actuator **622** and outlet-side valve actuator **624**, respectively, in accordance with some embodiments. Pump actuator **642**, inlet-side valve actuator **622**, and outlet-side valve actuator **624** may be disposed proximate to the back surface of the cassette recess **600**, and during operation, portions of pump actuator **642**, inlet-side valve actuator **622**, and outlet-side valve actuator **624** may extend beyond the back surface (e.g., during a forward stroke of the pump actuator **642** on piston **545** or during a forward stroke of inlet-side valve actuator **622** or outlet-side valve actuator **624** to close inlet-side valve **522** or outlet-side valve **524**, respectively).

(179) In accordance with some embodiments, motor **660** operatively coupled to the infusion pump system **10**, **11** (e.g., processing unit **12**, **13**). For example, processing unit **12**, **13** may provide signals to control a speed of the rotation of motor **660**. Motor **660** may be mechanically coupled to camshaft **662** such that a motor shaft is operative to rotate the camshaft **662**. In this regard, the camshaft **662** may be operatively coupled to motor shaft via gear disk **661**. However, in other embodiments, motor shaft and camshaft **662** may be one in the same.

(180) In certain embodiments, shaft **662** may include a plurality of cams **663a-c** for operating inlet-side valve actuator **622**, pump actuator **642**, and outlet-side valve actuator **624**. Each of cams **663a-c** may be formed and/or aligned with respect to the camshaft **662** to provide for proper valve and pump actuator operation as described herein. Accordingly, in certain embodiments, a single camshaft may be utilized in-line to operate both valve and pump actuators (e.g., reducing gearing requirements in comparison to some scotch-yoke pumping configurations). However, it is to be appreciated that other pump drive assemblies are contemplated with cassette **500** and cassette recess **600**, in accordance with the present disclosure.

(181) In some embodiments, sensor slot **628** may comprise an air-in-line detector **629**. Air-in-line detector **629** may be operable to detect air in the controllable fluid path of cassette **500** via fluid pathway extension member **528**, for example. Cassette recess **600** may include various mechanical couplings and operational interfaces, such as but not limited to an inlet-side pressure sensing probe within and/or proximate to inlet-side pressure sensing probe recess **632a** and an outlet-side pressure sensing probe within and/or proximate to outlet-side pressure sensing probe recess **634a** thereby enabling measurement of in-line pressure and fault isolation to a section of the fluid pathway. For example, inlet-side pressure sensing probe may operably contact upstream pressure dome **532** through a corresponding opening of interface-facing frame portion **516**. Similarly, outlet-side pressure sensing probe may operably contact downstream pressure dome **534** through a corresponding opening of frame portion **516**.

(182) With reference to the examples of FIGS. **8C**, **8D**, **9A**, and **9B**, the x-y positioning of cassette **500** within cassette recess **600** can be constrained by the pump structure **541** and pump structure recess **641a** mating interface, as well as fluid pathway extension member **528** and sensor slot **628** mating interface. In this regard, cassette **500** and cassette recess **600** can have two points of contact in the z-axis direction (e.g., an axis through and transverse to a general plane of the interface-facing surface of cassette body **510** of cassette **500** and cassette-facing surface of cassette recess **600**) for interlock alignment of the cassette with respect to the x-y positioning of the interface side of the cassette body **510**.

(183) It is to be understood that pump structure **541** may have various cross sectional geometries in addition to the circular cross-section support wall illustrated in the example of FIG. **8C** and other figures. For example, an external support wall of pump structure **541** may have a triangular, square, or other polygonal cross-section. Alternatively or in addition, the slider **570** may be configured have to tighter tolerances with respect to the x-axis and/or y-axis. For example, the x-axis tolerance

may be substantially tight so as to avoid or substantially limit any minor or miniscule rotation of the cassette body **510** during prolonged continuous operation with infusion pump system **10, 11**.

(184) During insertion of cassette **500**, once cassette body **510** is placed in cassette recess **600**, slider **570** can be longitudinally articulated to from a first position to a second position with respect to cassette body **510** (e.g., articulated downwardly in certain implementations; FIG. **9A**). In this second position cassette **500** will be latched within cassette recess **600** by virtue of protrusions **574** being engaged with slots **674**. In the second position, flow stop valve **564** is aligned under a stop valve guard **578** (e.g., ramped, rounded, or recessed surface) of interface-facing slider section **576** of slider **570** (FIGS. **8B** and **8F**). When the slider **570** is positioned in the second position, the portion of interface-facing slider section **576** does not contact flow stop valve **564** (or alternatively does not contact flow stop valve **564** sufficiently to activate flow stop valve **564**), and flow stop valve **564** operates to allow fluid to flow freely through flow stop valve **564** to outlet **514**.

(185) FIGS. **10A** and **10B** illustrate examples of cassette **700** within cassette recess **800** in accordance with certain embodiments. Cassette **700** and cassette recess **800** may have similar components and features as like numbered components and features in other example embodiments described herein.

(186) The plurality of protrusions **774** on slider **770** may be engaged with the plurality of slots **874** of cassette recess **800** such that cassette **700** is secured within cassette recess **800** for operation. Cassette recess **800** may be configured to receive the pump cassette such that each of the plurality of protrusions **774** may contact a respective flat face ramp portion **874a** of each of the plurality of cassette engagement slots **874**. The plurality of protrusions **774** may contact and slide along the respective flat face ramp portions **874a** to engage with the other portions of the cassette engagement slots **874** (e.g., the deeper portions of the L-channel). In some embodiments, each of the plurality of protrusion **774** may also comprise a flat face portion **774a** (FIG. **11A**) such that each flat face portion **774a** may contact and slide along the respective flat face ramp portion **874a**.

(187) With reference to the example cassette engagement positions of FIGS. **10A-10C**, other aspects of cassette **700** are provided with additional reference to the examples of FIGS. **11A** and **11B**. For example, flow stop valve **764** may be configured to restrict and/or regulate fluid flow proximal to outlet **714** of cassette body **710**. In some embodiments, slider **770** may be articulated from a first position (e.g., slider **770** oriented toward top of cassette body **710**) to a second position (e.g., slider **770** oriented toward bottom of cassette body **710**) after priming and/or staging procedures have been completed and cassette **700** is to be installed into cassette recess **800**. During insertion of cassette **700**, cassette body **710** may be generally aligned with cassette recess **800** and the flat face portions **774a** of the plurality of protrusions **774** may contact the respective flat face ramp portions **874a** of the cassette engagement slots **874**.

(188) As illustrated in the example of FIG. **10C**, slider **770** may be a lockable slider in the second position in accordance with certain embodiments. For example, slider **770** may include slider grip **772** having a flexible portion **782** that is configured to allow the slider to change from a locked state to an unlocked state along such that the slider **770** can articulate a longitudinal path. Flexible portion **782** may include a polygon tab **784** that interfaces with a body tab **786** disposed on a slider-facing side of cassette body **710** when the flexible portion **782** is in an unbiased state (e.g., not being squeezed or pinched by the user). For example, polygon tab **784** may have a square face that contacts a respective square face of body tab **786** when the slider **770** is in the second position such that the polygon tab **784** is aligned below the body tab **786**. Thus, to articulate slider **770** from the second position to the first position (e.g., articulating the slider **770** upwardly with respect to the cassette body **710**), flexible portion **782** must be biased (e.g., squeezed or pinched by the user), such that the polygon tab **784** is no longer aligned below nor is obstructed by the body tab **786**. Hence, the slider **770** can freely articulate upwardly to the first position. Flexible portion **782** may be capable of being biased by one or more slots disposed on the slider grip **772**, for example.

(189) Polygon tab **784** may include an angled face that is aligned above the body tab **786** when the

slider **770** is in the first position. Thus, when slider **770** is articulated from the first position to the second position (e.g., articulating the slider **770** downwardly with respect to the cassette body **710**), the polygon tab **784** will not be obstructed by the body tab **786** and the angled face of the polygon tab **784** will cause the polygon tab **784** to slide around the body tab and bias the flexible portion **782**. When the polygon tab **784** extends beyond the body tab while the slider **770** is being longitudinally articulated, the flexible portion **782** may snap back to the unbiased state and generate an audible sound (e.g., a click) and or vibration in the slider grip **772** when the flexible portion **782** contact the body tab **786**, for example, enabling the user or caregiver to infer that the slider **770** has reached the second position and is now locked.

(190) It is to be appreciated that, in other embodiments, slider **770** may be configured to be lockable at multiple lockable positions along a longitudinal path with respect to the rigid body for which the slider **770** articulates, such as but not limited to the first position and one or more intermediate positions. Such embodiments may be configured by variation of the size and/or shape of the polygon tab **784** and the body tab **786**, for example. However, other techniques for locking and/or biasing the various slider **770** and slider grip **772** embodiments may be used as would be understood given the benefit of the present disclosure.

(191) In accordance with certain embodiments, slider grip **772** may be generally elongate and longitudinally aligned parallel to the two opposing edge sections of cassette body **710**. For example, a width of the slider grip **772** may be less than a length of the slider grip **772**, and the slider grip **772** may be positioned in a relative center of cassette **700** to enhance balance and alignment features during the process of inserting the cassette **700** into cassette recess **800**.

(192) With reference to FIGS. **11A** and **11B**, slider **770** can be articulated between the first position and second position with respect to cassette body **710** such that flow stop valve **764** is actuated by one or more portions of the slider **770**. For example, flow stop valve **764** may be actuated by contact with interface-facing slider section **776** when the slider **770** is in the first position. Thus, fluid flow through the fluid pathway may be occluded in the first position. Flow stop valve **764** may be activated by contact with interface-facing slider section **776** when the slider **770** is in the first position. Thus, fluid flow through the fluid pathway may be occluded in the first position. When the slider **770** is positioned in the second position, a portion of interface-facing slider section **776** (e.g., stop valve guard **778**) does not contact flow stop valve **764** (or alternatively does not contact flow stop valve **764** sufficiently to activate flow stop valve **764**), and flow stop valve **764** operates to allow fluid to flow freely through flow stop valve **764** to outlet **714**.

(193) Furthermore, slider **770** may be articulated between the first position and the second position when cassette **700** is not engaged with cassette recess **800** with the aid of grip feature **790**. In certain embodiments, grip feature **790** may be arranged on an exterior surface of interface-facing frame portion **716** of cassette body **710**, and may be defined in part by a plurality ribs **792** extending from the exterior surface of the interface-facing portion **716**. However, in some embodiments, grip feature **790** may include a recessed portion extending into the exterior surface of interface-facing frame portion **716** (e.g., an indentation on the exterior surface that does not extend beyond a thickness of interface-facing frame portion **716**). Grip feature **790** may be generally oval-shaped indicative of a thumb or finger support for a user or caregiver, for example.

(194) In an example method, while holding cassette **700**, a user or caregiver may place a thumb of his or her first hand on grip feature **790** and longitudinally articulate, with one or more fingers of his or her first hand, for example, slider **770** of cassette **700** with respect to the cassette body **710** from a first position in which flow stop valve **764** occludes fluid flow to a second position in which flow stop valve **764** allows fluid flow. As noted herein, slider **770** of cassette **700** may be in an unlocked state in accordance with certain embodiments when the slider **770** is in the first position such that the user would not need to pinch the flexible portion **782** of the slider grip **772** (or otherwise apply a force other than a longitudinal force to the slider **770**) to articulate the slider **770** from the first position to the second position. In embodiments where the first position is lockable,

the flexible portion **782** of the slider grip **772** may be biased to unlock the slider **770** with the one or more finger of the first hand or one or more fingers and a thumb of the second hand, for example.

(195) FIGS. **12A** and **12B** illustrate examples of a disposable IV pump cassette **900** and corresponding cassette recess **1000** of an interface module. In accordance with certain embodiments, cassette **900** may comprise a cassette body **910** and a slider **970**. Cassette **900** may include certain may include certain visual indicators related to operation aspects of the cassette and the infusion pump system in general. For example, cassette may include identifiable images such as fluid drops indicating position of slider **970** for free-flow (e.g., flow stop valve **964** in an open position) and a patient figure proximal to outlet **914**. In accordance with some aspects, one or more cassette-seated sensors may be disposed within the cassette recess **1000** so as to inform central processing unit **12** that the cassette is locked or secured into place within the cassette recess **1000** or seat. For example, cassette recess may include a window **1004** (or aperture) such that cassette identifier **902** (FIG. **13A**) can be scanned. Cassette identifier **902** may include various information such as, but not limited to, a manufacturer, type, and use parameters of cassette **900**. Moreover, cassette identifier **902** may be disposed on a top half of the exterior surface of interface-facing frame portion **916** with respect to gravity during use. Thus, a bottom half of the exterior surface of interface-facing frame portion **916** can be reserved for pump drive assembly and flow stop valve features, in accordance with certain embodiments.

(196) Slider **970** can be fixably and slidably engaged with cassette body **910** such that slider **970** may articulate longitudinally with respect to cassette body **910**, but will be constrained within range of sliding motion such that the slider remains coupled to the cassette body **910**. Slider **970** may be formed from rigid plastic or polymer material having lubricating characteristics (e.g., incorporating silicon or polytetrafluoroethylene (PTFE) additives), and is clear or translucent in accordance with certain embodiment. In some embodiments, slider **970** may be polycarbonate. In accordance with certain aspects, slider **970** may be lockable at one or more positions, and may include a slider grip **972** for unlocking and articulating slider **970**. Slider **970** may also include a plurality of protrusion **974** or lugs that are configured to mate and be releasably lockable with a plurality of slots **1074** of the cassette recess **1000** (e.g., L-shaped locking channels).

(197) Each of the plurality of protrusion **974** may also comprise a flat face portion **974a** that is configured to interface with a respective flat face ramp portions **1074a** of the cassette engagement slots **1074**. In this regard, cassette **900** can be self-guided and self-latched into the cassette recess **1000**. Accordingly, a door or lever action is not required in order to retain the cassette **900** within the cassette recess **1000**.

(198) The plurality of protrusions **974** may be disposed at various locations on slider **970**. As illustrated in the examples of FIGS. **12C** and **13A**, slider **970** may comprise a first pair of protrusions **974** disposed on opposing edges proximal to a top of slider **970**, and a second pair of protrusions **974** disposed on opposing edges distal from the top of slider **970**. The first pair of protrusions **974** may be spaced apart a same distance as the second pair of protrusions **974** in certain embodiments. Cassette **900** may include a different top edge shape **952a** (e.g., generally straight) as compared to a bottom edge shape **952b** (e.g., arcuate or partially rounded). In this regard, cassette **900** orientation within cassette recess **1000** (along with non-vertically aligned inlet recess **1012** and outlet recess **1014**) is clear to a user or caregiver such that a cassette **900** is not inadvertently installed (or when being primed) in an inverted manner. Additionally, perimeter edges of cassette **900** and corresponding perimeter edges of cassette recess **1000** may include rounded corners and opposing and/or parallel straight longitudinal edges, in accordance with certain example aspects.

(199) Such perimeter edge features may provide benefit by enabling visual identification as well as prohibiting improper orientation or seating of cassette **900** with a corresponding cassette recess **1000**. Additionally, minimizing a size of cassette **900** and optimizing a surface area of one or more

corresponding cassette recess **1000** may be achieved using arcuate perimeter edge configurations. (200) Additionally, an overall size of cassette **900** and cassette recess **1000** may be reduced, in accordance with some aspects. For example, in certain embodiments, cassette body **910** may be extended longitudinally a length between 70 mm and 90 mm. For orientation reference with respect to the various views of the examples illustrated of FIGS. **12A** through **14E**, longitudinal axis or y-axis **195**, latitudinal axis or x-axis **196**, and depth axis or z-axis **197** are provided as a reference on certain figures (e.g., FIGS. **12D**, **13A**, and **14A-14D**).

(201) In this regard, depth aspects of cassette **900** is shown in the example of FIG. **12D**. For example, in certain embodiments, cassette body **910**, or a substantial portion thereof, may extend depth between 5 mm and 10 mm. Fluid pathway extension member **928** may further extend between 8 mm to 10 mm. In certain aspects, slider grip **972** may extend between 9 mm to 16 mm from cassette body **910**. It is to be appreciated that the process of cleaning of inlet recess **1012**, outlet recess **1014**, and cassette recess **1000** is made efficient in the shallow recess configuration in accordance with certain embodiments should any fluid or debris accumulate within cassette recess **1000**. The shallow recess configuration of cassette recess **1000**, and associated longitudinal alignment of cassette **900** such that a smaller of volumetric dimensions of cassette **900** (e.g., depth being smaller than length and width in certain embodiments) further enables additional space for arrangement of mechanical couplings and operational interfaces and optimizes the overall space requirements of cassette recess **1000** and infusion pump system in general.

(202) Various types, placement, and orientations of the plurality of protrusions **974** disposed on slider **970** are contemplated in the present disclosure. Aspects of the various cassette-coupling techniques illustrated in the example cassette embodiments **100**, **300**, **500**, **700** described herein may be further combined and arranged into additional configurations suitable for specific implementations given the benefit of the present disclosure.

(203) Moreover, in accordance with certain aspects, features of cassette recess **1000** are designed to avoid wear down and/or risk of malfunction. For example, the plurality of slots **1074** arranged within cassette recess **1000** may be devoid of any movable latching mechanism in certain embodiments as such movable latching mechanisms may be susceptible to excessive wear and mechanical failure over repeated use with multiple disposable IV cassettes **900**.

(204) In operation, cassette **900** can be loaded directly into cassette recess **1000**. In this regard, the direct loading of the cassette **900** will enable avoidance of shear forces that might otherwise be applied to the sensors, alignment features, and other engaging interfaces of cassette-facing surface **1016** of cassette recess **1000** from interaction with the interface-facing side of cassette body **910** as it is loaded into cassette recess **1000**. It is to be understood that modification to the various features of cassette **900** can be made to accommodate the various cassette-coupling techniques disclosed herein.

(205) Cassette body **910** may comprise interface-facing frame portion **916** and slider-facing base portion **919** (FIGS. **12E**, **13A**, and **13B**) with membrane **917** disposed substantially therebetween. Portions of membrane **917** may extend through or be accessible from some openings of frame portion **916** (e.g., upstream pressure dome **932**, downstream pressure dome **934**, inlet-side valve **922**, and outlet-side valve **924**). In accordance with certain embodiments, membrane **917** can be a compliant material co-molded to the frame portion **916** and sealingly engaged with base portion **919** for defining a fluid pathway through cassette body **910** from inlet **912** to outlet **914**. Mating edges of frame portion **916** and base portion **919** may be connected by fusing, welding, gluing, or the like. Membrane **917** and base portion **919** may further define a plurality of other features, some of which may be accessed through openings in frame portion **916**.

(206) Frame portion **916**, membrane **917**, and/or base portion **919** may define features in or along the fluid pathway, in accordance with certain embodiments. For example, beginning from inlet **912**, the fluid pathway may include features such as, but not limited to, upstream pressure dome **932** (e.g., an inlet-side compliant reservoir), inlet-side valve **922**, pump chamber **925**, outlet-side valve

924, downstream pressure dome **934** (e.g., an outlet-side compliant reservoir), fluid pathway extension member **928**, and flow stop valve **964**. Other features that are not in or along the fluid pathway, but are disposed on cassette body **910**, may include positioning port **920** and slider stopper **951**. With respect to extension member **928**, a portion of the fluid pathway can be extended away or protrude orthogonally from the generally flat and planar exterior surface of interface-facing frame portion **916** so as to make the fluid in the fluid pathway available for certain detection techniques performed by infusion pump system **10**, **11**. As illustrated in the example of FIG. **12D**, for example, fluid pathway extension member **928** may be formed from orthogonally extending portions of frame portion **916**, membrane **917**, and/or base portion **919**.

(207) With reference to the examples of FIGS. **12C**, **12D**, and **12E** (illustrating an enlarged, cross-sectional perspective view of cassette **900**), slider **970** may be a lockable slider in a first and second position in accordance with certain embodiments. Thus, slider **970** may be configured to articulate such that the slider **970** is movably fixable at one or more positions with respect to cassette body **910**. For example, slider **970** may include slider grip **972** having a flexible portion **975** that is configured to allow the slider to change from a locked state to an unlocked state such that the slider **970** can articulate along a longitudinal path. The longitudinal path of slider **970** may be constrained by one or more slider stoppers, for example top slider stopper **951a** and bottom slider stopper **951b**. Flexible portion **975** may include a slider grip tab **966** that interfaces with a body tab **968** disposed on a slider-facing side of cassette body **910** when the flexible portion **975** is in an unbiased state (e.g., not being squeezed or pinched by the user). For example, slider grip tab **966** may have a square face that contacts a respective square face of body tab **968** when the slider **970** is in the second position such that the slider grip tab **966** is aligned below the body tab **968**. Thus, to articulate slider **970** from the second position to the first position (e.g., articulating the slider **970** upwardly with respect to the cassette body **910**), flexible portion **975** must be biased (e.g., squeezed or pinched by the user), such that the slider grip tab **966** is no longer aligned below nor is obstructed by the body tab **968**. Hence, the slider **970** can freely articulate upwardly to the first position. Flexible portion **975** may be capable of being biased by one or more slots disposed on the slider grip **972**, for example.

(208) Slider grip tab **966** may also include a square face that is aligned above the body tab **968** when the slider **970** is in the first position. Thus, to articulate slider **970** from the first position to the second position (e.g., articulating the slider **970** downwardly with respect to the cassette body **910**), the flexible portion **975** must be biased (e.g., squeezed or pinched by the user), such that the slider grip tab **966** is no longer aligned above nor is obstructed by the body tab **968**. Hence, the slider **970** can freely articulate downwardly to the second position. In some embodiments, when the slider grip tab **966** extends beyond the body tab while the slider **970** is being longitudinally articulated, and the user relates pressure to the flexible portion **975** (e.g., whereby the slider grip tab **966** and the body tab **968** are parallel aligned and contacting each other on respective sides), the flexible portion **975** may snap back to the unbiased state and generate an audible sound (e.g., a click) and/or vibration in the slider grip **972** when the flexible portion **975** contact the body tab **968**. In this regard, the user or caregiver can infer that the slider **970** has reached the first or second position and is now locked.

(209) It is to be appreciated that, in other embodiments, slider **970** may be configured to be lockable at multiple lockable positions along a longitudinal path with respect to the rigid body for which the slider **970** articulates, such as but not limited to one or more intermediate positions. Such embodiments may be configured by variation of the size and/or shape of the slider grip tab **966** and the body tab **968**, for example. However, other techniques for locking and/or biasing the various slider **970** and slider grip **972** embodiments may be used as would be understood given the benefit of the present disclosure.

(210) In accordance with certain embodiments, slider grip **972** may be generally elongate and longitudinally aligned parallel to the two opposing edge sections of cassette body **910**. For

example, a width of the slider grip **972** may be less than a length of the slider grip **972**, and the slider grip **972** may be positioned in a relative center of cassette **900** (e.g., particularly when locked in the second position) to enhance balance and alignment features during the process of inserting the cassette **900** into cassette recess **1000**.

(211) In accordance with certain embodiments, membrane **917** may be formed from a thermoplastic elastomer (TPE). Characteristics of certain TPEs can enable effective co-molding with other materials, for example, polycarbonate. Accordingly, in some embodiments, membrane **917** may be co-molded to frame portion **916** and a striker may be co-molded to a portion of membrane **917** defining a flow stop valve **964**. However, in some embodiments, membrane **917** can be formed from silicon, a silicon-based compound, an elastomeric material suitably compliant for fluid flow, or the like.

(212) In accordance with certain embodiments, interface-facing frame portion **916** and slider-facing base portion **919** may be formed from a rigid plastic such as, but not limited to, a polycarbonate. Additionally, the rigid plastic of frame portion **916** and base portion **919** may be clear or translucent. The material of membrane **917** (e.g., TPE or other compliant material) and rigid plastic slider **970** may also be clear or translucent, thereby allowing a user or caregiver to readily observe fluid passage through a substantial portion of the fluid pathway of cassette body **910**. In some embodiments, the fluid pathway portion of cassette body **910** will be clear or translucent, and other portions will be frosted so as to direct a user or caregiver's attention to the fluid pathway.

(213) In some implementations, slider **970**, base portion **919**, and membrane **917** may be clear or translucent (or at least some portions along the fluid pathway), and the frame portion **916** may not be translucent. For example, the frame portion **916** may be colored in a manner so as to contrast against a color or tint of the fluid expected to be used with cassette **900**. In some embodiments, a lens area may be disposed on base portion **919** and/or slider **970**.

(214) FIGS. **13C-13E** are enlarged, longitudinal cross-sectional views of cassette body **910** proximate to pump chamber **925**. Opening **925a** of the pump chamber **925** (e.g., when piston **945** is retracted) is disposed between the inlet-side valve **922** and outlet-side valve **924** along a bottom the fluid pathway section **923**. In some embodiments, the nose or tip of the piston **945** is substantially aligned with a bottom of fluid pathway section **923** (FIG. **13E**) to eliminate any dead space (e.g., internal space where air may accumulate) in pump chamber **925** as well as reduce any drag associated with the fluid flow through the fluid pathway section **923**. Thus, in some embodiments, the dead space is less than 1% of the total volume of the pump chamber **925**. For example, in certain embodiments, the volume of the pump chamber is 80 microliters when the piston **945** is fully retracted in its reciprocating cycle.

(215) In this regard, it can be advantageous to place the inlet-side valve **922** and outlet-side valve **924** close together along the fluid pathway section **923** proximal to the pump chamber **925**. For example, a distance between the inlet-side valve **922** and outlet-side valve **924** is approximately between 4 millimeters and 7 millimeters in some embodiments. It is to be appreciated that piston pump techniques can provide repeatedly precise positive displacement of fluid in the pump chamber **925**.

(216) In accordance with certain embodiments, a section or tract of the fluid pathway leading from the area of the outlet-side valve **924** may comprise a straight edge portion **923a** that is tangent to an arcuate edge **934a** of the downstream pressure dome **934**. The tangentially aligned straight edge **923a** and arcuate edge **934a** are top edge portions with respect to an orientation of the cassette **900** with respect to gravity, for example, as the cassette **900** would be installed into cassette recess **1000** such that longitudinal axis or y-axis **195** is substantially aligned with gravity.

(217) With respect to the orientation of pump chamber **925** of cassette **900** and pump chamber having pump chamber opening/access **125** of cassette **100**, in certain embodiments, it may be advantageous to have pump chamber **925** in order to prevent or limit the impact of air bubbles in pump chamber accuracy. For example, in pump chamber having pump chamber opening/access

125 during the delivery phase of the pump cycle, fluid will be expelled first and any air that accumulates in the pump chamber of cassette **100** and between the inlet-side valve **122**, outlet-side valve **124** will remain thereby decreasing pumping accuracy of the system. In contrast, pump chamber **925** of cassette **900** will first expel any air that is in the pump chamber, thereby preventing air from accumulating in the pump chamber **925** and in the region between the inlet-side valve **922**, outlet-side valve **924** and maintaining accuracy. For example, with additional reference to the example of FIG. **13F**, one or more fluid sensors may be disposed within sensor slot **1028**. The one or more fluid sensors disposed within sensor slot **1028** can be ultrasonic sensors configured as an air-in-line detector, for example. In certain embodiments, extension member **928** may be disposed on cassette body **910** and positioned along the fluid pathway between downstream pressure dome **934** and flow stop valve **964**. However, in some embodiments, extension member **928** can be positioned at other locations along the fluid pathway such as, but not limited to, between inlet **912** and upstream pressure dome **932**. Additionally, in other embodiments, a plurality of extension members **928** with a plurality of corresponding sensor slots **228** may be positioned along a fluid pathway of cassette body **910**.

(218) With reference to the examples illustrated in FIGS. **13A-13E**, cassette body **910** may include a pump drive assembly in accordance with certain embodiments. For example, the pump drive assembly may include pump drive mechanism **942** for receiving pump actuator **1042** of cassette recess **1000**. Pump drive mechanism **942** can be operatively coupled to piston **945** slidably engaged within piston guide **943** or casing (e.g., generally cylindrical or frustoconical casing) such that reciprocal movement of piston **945** may urge fluid into and out of the pump chamber **925** and through the fluid pathway of cassette body **910**. In this regard, the pump chamber **925** may be defined by a portion of the piston guide **943** or casing distal from the pump drive mechanism **942** that is adjacent to and fluidly coupled with a tract or section of the fluid pathway between inlet-side valve **922** and outlet-side valve **924**. Piston **945** may comprise one or more one or more slideable seals **946** for defining a changeable volume portion of the pump chamber during reciprocal movement of piston **945** slidably disposed within the piston guide **943**. Thus, the volume in pump chamber **925** changes with the reciprocal motion of the piston **945** such that a volume of the pump chamber **925** may be varied by movement of the piston **945** in accordance with certain embodiments.

(219) A wiper seal (not shown) may be positioned within or proximal to piston guide **943** and slidably engaged with piston **945** thereby reducing the possibility of any substances (e.g., dirt or dried fluid particles) near the cassette **900** from contacting one or more slideable seals of the piston **945**. The one or more slideable seals **946** of the piston **945** may contact an internal wall piston guide **943** to form a movable barrier of the pump chamber **925**. Additionally, piston **945** may include a reduced tip portion for more precise volumetric displacement of fluid into and out of pump chamber **925**.

(220) For example, pumping operation of infusion pump system **10, 11** when cassette **900** is primed and seated in cassette recess **1000** may comprise activating outlet-side valve actuator **1024** such that outlet-side valve **924** is closed or sealed while activating inlet-side valve actuator **1022** such that inlet-side valve **922** is opened. Opening of inlet-side valve **922** may coincide with or occur shortly after a reverse stroke of piston **945** (e.g., a movement of piston **945** away from pump chamber **925**). Accordingly, fluid can flow from upstream pressure dome **932** to pump chamber **925**. Alternatively, or in addition to, outlet-side valve **924** may comprise a one-way valve mechanism that permits flow of fluid under normal conditions in one direction (from a fluid container to a patient). Additionally, inlet-side valve **922** may also comprise a one-way valve mechanism permitting flow of fluid in one direction (from a fluid container to a patient) under normal operating conditions. In this configuration, cassette recess **1000** may not need to incorporate either outlet-side valve actuator **1024** or inlet-side valve actuator **1022**. Outlet-side valve **924** and inlet-side valve **922** may limit flow of fluid in one direction, but permit flow in an

opposite direction in the event fluid pressure overcomes a cracking pressure of the valves.

(221) Continuing with the valve-operated implementation, pumping operation may comprise activating outlet-side valve actuator **1024** such that outlet-side valve **924** is open while activating inlet-side valve actuator **1022** such that inlet-side valve **922** is closed or sealed. Opening of outlet-side valve **924** may coincide with or occur shortly before a forward stroke of piston **945** (e.g., a movement of piston **945** toward the opening of the pump chamber **925** such that the volume of the pump chamber **925** is reduced). Thus, fluid can flow from pump chamber **925** to downstream pressure dome **934** and consequently urging fluid out outlet **914**.

(222) In certain embodiments, pump chamber **925** is a smaller volume than one or both of upstream pressure dome **932** and downstream pressure dome **934**. Accordingly, larger and compliant upstream pressure dome **932** and/or downstream pressure dome **934** can address any backpressure issues in the IV set, thereby allowing for an accurate and precise volume of fluid entering pump chamber **925** to be pumped.

(223) Referring to FIGS. **13A-13F**, pump drive mechanism **942** and pump actuator **1042** may be configured as a reciprocating motion mechanism (e.g., a scotch-yoke configuration) in certain implementations. Pump drive mechanism **942** may be accessible by pump actuator **1042** via an aperture through interface-facing side section **976**. In such implementations, pump drive mechanism **942** may include opposing ramp portions for guiding a rotatable pin of pump actuator **1042** toward an elongate slot of pump drive mechanism **942**. For example, the outer edges of the opposing ramp portions may be arranged a distance that will ensure engagement with the rotatable pin of pump actuator **1042**. When the rotatable pin contacts one of the ramp portions, the pump drive mechanism **942** will move to align the elongate slot of pump drive mechanism **942** with the with the rotatable pin of pump actuator **1042**. However, it is to be appreciated that other pump drive assemblies are contemplated with cassette **900** and cassette recess **1000** in accordance with the present disclosure.

(224) In certain embodiments, cassette recess **1000** may include an upstream pressure sensing probe **1032** and downstream pressure sensing probe **1034** enabling measurement of in-line pressure and fault isolation to a section of the fluid pathway. For example, upstream pressure sensing probe **1032** may operably contact upstream pressure dome **932** through a corresponding opening of interface-facing frame portion **916**. Similarly, downstream pressure sensing probe **1034** may operably contact downstream pressure dome **934** through a corresponding opening of frame portion **916**.

(225) The x-y positioning of cassette **900** within cassette recess **1000** can be constrained by the positioning port **920** and positioning protrusion **1020** mating interface, as well as fluid pathway extension member **928** and sensor slot **1028** mating interface (e.g., air-in-line detector feature). In this regard, cassette **900** and cassette recess **1000** can have two points of contact in the z-axis direction (e.g., an axis through and transverse to a general plane of the interface-facing surface of cassette body **910** of cassette **900** and cassette-facing surface of cassette recess **1000**) for interlock alignment of the cassette with respect to the x-y positioning of the interface side of the cassette body **910**.

(226) According to certain aspects, positioning port **920** may be located proximal to inlet-side valve **922** and outlet-side valve **924**, and correspondingly mating positioning protrusion **1020** may be located proximal to inlet-side valve actuator **1022** and outlet-side valve actuator **1024**. In this regard, the cassette **900** may be positioned properly without over constraining the cassette **900**.

(227) In other words, the controlling index for proper x-axis and y-axis alignment of the interface-facing frame portion **916** is the positioning port **920** and positioning protrusion **1020** mating interface, in certain embodiments. In some embodiments, the positioning port **920** and positioning protrusion **1020** mating interface, and fluid pathway extension member **928** and sensor slot **1028** mating interface, together form the controlling index for proper x-axis and y-axis alignment of the interface-facing frame portion **916** (e.g., dual-feature x-y positioning scheme).

(228) However, in certain embodiment, the latching engagement of slider **970** with cassette recess **1000** provides tight tolerances for z-axis positioning of the cassette body **910** and corresponding interface-facing frame portion **916** for operation with the cassette-facing surface **1016** of cassette recess **1000** (e.g., z-axis alignment or distances require for proper operation of the pressure sensors as well as valve actuators). It is to be understood that techniques of z-axis alignment or depth other than slider protrusion **974** positioning are contemplated.

(229) It is to be understood that positioning protrusion **1020** may have various cross sectional geometries in addition to the circular cross-section protrusion/pin-type illustrated in FIG. **13F** and other figures. For example, pin may have a triangular, square, or other polygonal cross-section. For instance, a polygonal shape may be used to provide tighter tolerances to the x-y positioning of the cassette body **910** when cassette **900** has an air-in-line features the does not protrude into and received by cassette recess **1000**. Alternatively or in addition, the slider **970** may be configured have to tighter tolerances with respect to the x-axis and/or y-axis. For example, the x-axis tolerance may be substantially tight so as to avoid or substantially limit any minor or miniscule rotation of the cassette body **910** during prolonged continuous operation with infusion pump system **10, 11**. Moreover, the positioning port **920** and positioning protrusion **1020** mating interface can be reversed in some implementation (e.g., the positioning protrusion may be disposed on cassette **900**).

(230) However, in some implementations of cassette **900** and cassette recess **1000**, a dual-feature x-y positioning scheme as described herein can be advantageous so as to avoid too much stress on a single mating feature. For example, positioning protrusion **1020** having a circular cross-section cooperatively engaging with positioning port **920** can provide tight tolerances for either the x-axis or y-axis alignment of interface-facing frame portion **916**, but not necessarily both x-axis or y-axis alignment as rotation between positioning protrusion **1020** and positioning port **920** can occur. However, in conjunction with a second x-y positioning such as fluid pathway extension member **928** and sensor slot **228** mating interface, tight tolerances can be achieved without undue stress on either of the x-y positioning features, as well as the cassette **900** and cassette recess **1000** in general.

(231) Accordingly, advantages of the x-y positioning schemes disclosed herein include, but are not limited to, preventing sheering to sensor features (e.g., upstream pressure sensing probe **1032** and downstream pressure sensing probe **1034** may not engage with sensors until they are properly aligned by the x-y positioning features, aiding in full and proper valve operation (e.g., inlet-side valve actuator **1022** and outlet-side valve actuator **1024** may not fully close a corresponding valve if misaligned on the x-axis and/or y-axis occurs), and assisting with communicating a correct timing for latching slider **970** of cassette **900** with cassette recess **1000** (e.g., caregiver can sense engagement of x-y positioning features between cassette **900** and cassette recess **1000**).

(232) FIGS. **14A-4D** illustrate an example of a cassette engagement sequence with cassette **900** and cassette recess **1000**. Cassette **900** may be aligned such that the plurality of protrusions **974** on slider **970** may be aligned along z-axis **197** for engagement with the plurality of slots **1074** of cassette recess **1000**. In accordance with certain aspects, cassette **900** may have a longitudinal length along y-axis **195**, a lateral width along x-axis **196**, and a depth along z-axis **197**. As illustrated in FIG. **4A** and described herein, the depth of the cassette **900** may be a smaller dimension than either the length or the width of cassette **900**. In this regard, cassette **900** is front loaded into cassette recess **1000**.

(233) As illustrated in FIGS. **14A** and **14B**, the plurality of protrusions **974** on slider **970** of cassette **900** may engage with the plurality of slots **1074** of cassette recess **1000** such that cassette body **910** may be generally aligned with cassette recess **100** and the flat face portions **974a** of the plurality of protrusions **974** may contact the respective flat face ramp portions **1074a** of the cassette engagement slots **1074** (e.g., slider **970** in the second position with respect to cassette body **910**). In certain embodiments, the flat face portions **974a** of the plurality of protrusions **974** may have an

angled plane different from a general plane of the interface-facing surface of cassette body **910** of cassette **900** and cassette-facing surface **1016** of cassette recess **1000**.

(234) Next, as illustrated in FIG. **14C**, when a force is into the to the slider **970** substantially orthogonal to general plane of the interface-facing surface of cassette body **910** of cassette **900** and cassette-facing surface **1016** of cassette recess **1000**, the slider **970** articulates in a direction from the second position to the first position. In this regard, the plurality of engagement protrusions **974** may slide along the respective flat face ramp portions **1074a** of the cassette engagement slots **1074** deeper into the cassette recess **1000**.

(235) Next, as illustrated in FIG. **14D**, slider **970** may be longitudinally articulated along y-axis **195** such that cassette **900** is latched and locked by slider **970** within cassette recess **1000**. For example, the plurality of engagement protrusions **974** may be positioned behind the flat face ramp portions **1074a** of the cassette engagement slots **1074** while cassette **900** is engaged and operational with cassette recess **1000**. It is to be noted that in some embodiments, loading of cassette **900** within cassette recess **1000** may be laterally oriented (e.g., 90° rotation of z-axis **197** so that x-axis **196** and y-axis **195** are switched).

(236) In certain embodiments, flow stop valve **964** may be configured to restrict and/or regulate fluid flow proximal to outlet **914** of cassette body **910**. In the first position, flow stop valve **964** is aligned under a portion (e.g., flat surface) of interface-facing slider section **976** of slider **970**. When the slider **970** is positioned in the first position, the portion of interface-facing slider section **976** contacts and activates flow stop valve **964** such that fluid flow is occluded at that position of the fluid pathway proximal to outlet **914** cassette body **910**. Therefore, fluid leakage can be avoided during the final preparation stages (e.g., after priming of cassette **900**) and prior to the start of insertion stage into cassette recess **1000**, in accordance with certain implementations. Moreover, in accordance with certain configurations, the first position of slider **970** may correspond to a position of cassette **900** for disengaging with cassette recess **1000**.

(237) Slider **970** can be articulated to a second position with respect to cassette body **910** (e.g., articulated downwardly in certain implementations). In this second position, cassette **900** will be latched within cassette recess **1000** by virtue of protrusions **974** being engaged with slots **1074** and locked by virtue of slider **970** with respect to cassette body **910**. In the second position, flow stop valve **964** is aligned under a stop valve guard **978** (e.g., ramped, rounded, or recessed surface) of interface-facing slider section **976** of slider **970**. When the slider **970** is positioned in the second position, the portion of interface-facing slider section **976** does not contact flow stop valve **964** (or does not contact flow stop valve **964** sufficiently to activate flow stop valve **964**), and flow stop valve **964** operates to allow fluid to flow freely through flow stop valve **964** to outlet **914**.

(238) In accordance with certain embodiments, stop valve guard **978** may be positioned underneath a portion of and/or proximal to an edge along interface-facing slider section **976** of slider **970** such that when cassette **900** is securely latched or locked within cassette recess **1000** (e.g., in the second position), stop valve guard **978** is positioned above flow stop valve **964**. In this regard, stop valve guard **978** can protect flow stop valve **964** from being inadvertently depressed and activated to restrict fluid flow while cassette **900** is in use. For example, a force applied to the slider side of cassette **900** while locked within cassette recess **1000** would not depress flow stop valve **964** as the lateral tolerances of the slidable coupling between cassette **900** and slider **970** are tighter than a distance between the inner surface of the stop valve guard **978** and an outer surface of the flow stop valve **964**. In this regard, distances between stop valve guard **978** and flow stop valve **964** may be optimized such that anticipated forces applied to cassette **900** (e.g., from a user or caregiver inadvertently bumping cassette **900** or road vibrations in moving ambulance setting) may not cause an undesired activation of flow stop valve **964**.

(239) Similarly, when cassette **900** is to be disengaged from cassette recess **1000**, slider **970** can be unlatched and/or unlocked from cassette recess **1000** by accessing grip **972**, squeezing flexible portion **975** to unlock slider **970**, and articulating slider **970** back to the first position with respect

to cassette body **910**. Some amount of force may be required by the user to articulate slider **970** to the first position as the plurality of protrusions **974** may be securely latched with corresponding slots **1074** while the slider **970** is in the second position. Once slider **970** is in the first position, cassette **900** may be removed from cassette recess **1000** by pulling grip **972** outwardly.

(240) Furthermore, slider **970** may be articulated from the first position to the second position when cassette **900** is not engaged with cassette recess **1000**. Thus, flow stop valve **964** will operate to allow fluid to flow freely through flow stop valve **964** to outlet **914**. Therefore, cassette **900** may be primed with fluid while being disengaged from cassette recess **1000** when slider **970** is in the second position (but not latched with cassette recess **1000**).

(241) It is to be understood that in other implementations, the stop valve guard feature can be a recess distal from an edge of interface-facing slider section **976**, or an aperture or slit extending through interface-facing slider section **976**, for example. Moreover, in some embodiments, flow stop valve and stop valve guard features may be positioned on the slider-facing side of cassette body **910**. It is to be understood that various constructions of flow stop valve **964** are contemplated in the present disclosure as described herein. As illustrated in FIGS. **13F** and **14E**, cassette recess **1000** may include interface structures operable in conjunction with infusion pump systems **10**, **11**. For example, pump actuator **1042** may be configured to provide a rotational force aligned with piston drive mechanism **942** (e.g., for use with scotch-yoke pumping configurations).

(242) Cassette recess **1000** additionally includes inlet-side valve actuator **1022** and outlet-side valve actuator **1024**, in accordance with some embodiments. Inlet-side valve actuator **1022** and outlet-side valve actuator **1024** may be disposed proximate to the back surface of the cassette recess **1000**, and during operation, portions of inlet-side valve actuator **1022** and outlet-side valve actuator **1024** may extend beyond the back surface (e.g., during a forward stroke of inlet-side valve actuator **1022** or outlet-side valve actuator **1024** to close inlet-side valve **922** or outlet-side valve **924**, respectively).

(243) In accordance with some embodiments, motor **1060** operatively coupled to the infusion pump system **10**, **11** (e.g., processing unit **12**, **13**). For example, processing unit **12**, **13** may provide signals to control a speed of the rotation of motor **1060**. Motor **1060** may be mechanically coupled to a plurality of gears **1061** such that a motor shaft is operative to rotate pump actuator **1042** as well as reciprocate inlet-side valve actuator **1022** and outlet-side valve actuator **1024** via a cam arrangement **1063**. Cam arrangement **1063** may comprise a plurality of cams formed and/or aligned to provide proper valve and pump actuator operation as described herein. An optical flag detector **1065** may be included to identify rotations of the cam arrangement, for example. However, it is to be appreciated that other pump drive assemblies are contemplated with cassette **900** and cassette recess **1000**, in accordance with the present disclosure.

(244) In some embodiments, scanner (or reader) **1002** may be operably coupled to infusion pump system **10**, **11** (e.g., processing unit **12**, **13**) such that cassette identifier **902** can be scanned through window (or aperture) **1004**. Additionally, an air-in-line detector **1029** may be operable to detect air in the controllable fluid path of cassette **900** via fluid pathway extension member **928**, for example. Cassette recess **1000** may include various mechanical couplings and operational interfaces, such as but not limited to an inlet-side pressure sensing probe **1032** and an outlet-side pressure sensing probe recess **1034** thereby enabling measurement of in-line pressure and fault isolation to a section of the fluid pathway. For example, inlet-side pressure sensing probe **1032** may operably contact upstream pressure dome **932** through a corresponding opening of interface-facing frame portion **916**. Similarly, outlet-side pressure sensing probe may operably contact downstream pressure dome **934** through a corresponding opening of frame portion **916**.

(245) FIGS. **15A-15C** illustrate examples of a disposable IV pump cassette **1100** and corresponding cassette recess **1200** of an interface module. In accordance with certain embodiments, cassette **1100** may comprise a cassette body **1110** and a slider **1170**. Cassette **1100** and cassette recess **1200** may have similar components and features as like numbered components and features in other example

embodiments described herein, particularly with respect to cassette **900** and cassette recess **1000**.

(246) With reference to the examples of FIGS. **15A** and **15B** (illustrating an enlarged, cross-sectional perspective view of cassette **1100**), slider **1170** may be a lockable slider in a first and second position in accordance with certain embodiments. Thus, slider **1170** may be configured to articulate such that the slider **1170** is movably fixable at one or more positions with respect to cassette body **1110**. For example, slider **1170** may include slider grip **1172** such that a user or caregiver may hold to articulate slider **1106**. Slider **1170** may also include a flexible segment **1175** that is configured to retain the slider **1170** in a fixed position with respect to the cassette body. In this regard, the flexible segment **1175** may provide an amount of resistance or friction to moving or articulating slider **1170** along a longitudinal path.

(247) The longitudinal path of slider **1170** may be constrained by one or more slider stoppers, for example top slider stopper **1151a** and bottom slider stopper **1151b**. Flexible segment **1175** may be capable or being biased (e.g., bent or bowed) to overcome the resistance provided for restricting longitudinal motion. For example, flexible segment **1175** may include one or more slots disposed on a portion of the slider **1170** facing the cassette body to alter the pliability of that portion.

(248) In certain embodiments, flexible segment **1175** may include a slider segment tab **1166** that interfaces with a body tab **1168** disposed on a slider-facing side of cassette body **1110**. For example, slider segment tab **1166** may have an angled face (e.g., a triangular cross-section) that contacts a respective angled face of an edge portion of body tab **1168** when the slider **1170** is in the second position such that the slider segment tab **1166** is aligned longitudinally below the body tab **1168**. Thus, to articulate slider **1170** from the second position to the first position (e.g., articulating the slider **1170** upwardly with respect to the cassette body **1110**), a user or caregiver must overcome the resistance provided by the flexible segment **1175** for fixing the slider **1170** in the second position. When sufficient force is applied to the slider **1170**, slider flexible segment **1166** will bend or bow and slider segment tab **1166** will slide over a top of body tab **1168** until slider segment tab **1166** reaches an opposite side of body tab **1168**, for example. The slider **1170** will then be in the first position with respect to cassette body **1110**, in accordance with certain embodiments.

(249) Slider segment tab **1166** may have a similarly angled face that contacts a respective angled face of an edge portion of body tab **1168** when the slider **1170** is in the first position such that the slider segment tab **1166** is aligned longitudinally above the body tab **1168**. Thus, to articulate slider **1170** from the first position back to the second position (e.g., articulating the slider **1170** downwardly with respect to the cassette body **1110**), the user or caregiver must again overcome the resistance provided by the flexible segment **1175** for fixing the slider **1170** in the first position. When sufficient force is applied to the slider **1170**, slider flexible segment **1166** will bend or bow and slider segment tab **1166** will slide over a top of body tab **1168** until slider segment tab **1166** reaches an opposite side of body tab **1168**, for example. The slider **1170** will then be in the second position with respect to cassette body **1110**, in accordance with certain embodiments.

(250) When the slider segment tab of the flexible segment **1175** reaches either end of the body tab **1168**, the flexible segment may snap back to the unbiased state (e.g., unbent or unbowed position) and generate an audible sound (e.g., a click) and/or vibration in the slider grip **1172** when the flexible segment **1175** contact the top of the body tab **1168**. In this regard, the user or caregiver can infer that the slider **1170** has reached the first or second position and is now fixed in position. It is to be understood that the amount of force required by a user to articulate slider **1170** to and from the one or more position may adapted to various implementations of cassette **1100** by altering characteristics and geometries of the flexible segment **1175**, slider segment tab **1166**, and/or body tab **1168**.

(251) With reference to the example of FIG. **15C**, slider **1170** may be articulated between the first position and the second position when cassette **1100** is not engaged with cassette recess **1200** with the aid of grip feature **1190**. In certain embodiments, grip feature **1190** may be arranged on an exterior surface of interface-facing frame portion **1116** of cassette body **1110**, and may be defined

in part by a plurality of ribs **1192** extending from the exterior surface of the interface-facing portion **1116**. Grip feature **1190** may include a recessed portion extending into the exterior surface of interface-facing frame portion **1116** (e.g., an indentation on the exterior surface that does not extend beyond a thickness of interface-facing frame portion **1116**). Grip feature **1190** may be generally oval-shaped indicative of a thumb or finger support for a user or caregiver, for example.

(252) For example, while holding cassette **1100**, a user or caregiver may place a thumb of his or her first hand on grip feature **1190** and longitudinally articulate, with one or more fingers of his or her first hand, for example, slider **1170** of cassette **1100** with respect to the cassette body **1110** from a first position in which flow stop valve **1164** occludes fluid flow to a second position in which flow stop valve **1164** allows fluid flow. As noted herein, slider **1170** of cassette **1100** may be in a movably fixed position when the slider **1170** is in either the first position or the second position such that the user would not need to apply a longitudinal force to overcome the resistance or friction provided by flexible segment **1175** (e.g., by the interaction between segment tab **1166** and body tab **1168**).

(253) It is to be understood that aspects of the various embodiments disclosed herein may be intermixed and incorporated such that alternative embodiments may include one or more aspects from one or more embodiments combined in another embodiment. For example, and without limiting the scope of such combinations, alternative embodiments may include: flat face ramp portions **874a**, **1074a**, **1274a** of cassette recess **800**, **1000**, and **1200** with any of the other cassette recesses **200**, **400**, **600**; flow stop valve **164** and interface-facing slider section **176** include one or more cassette-facing detents **179** of cassette **100** (FIGS. 5E and 5D) with any of the other cassettes **300**, **500**, **700**, **900**, **1100**; vertically-oriented and lockable slider grips **772**, **972** of cassettes **700**, **900** with any of the other cassettes **100**, **300**, **500**, **1100**; and pump structure **541** configured for orthogonal piston assembly operation of cassette **500** with any of the other cassettes **100**, **300**, **700**, **900**, **1100**.

(254) The subject technology is illustrated, for example, according to various aspects described above. Various examples of these aspects are described as numbered concepts or clauses (1, 2, 3, etc.) for convenience. These concepts or clauses are provided as examples and do not limit the subject technology. It is noted that any of the dependent concepts may be combined in any combination with each other or one or more other independent concepts, to form an independent concept. The following is a non-limiting summary of some concepts presented herein:

(255) Concept 1. A pump cassette comprising: a rigid body comprising a frame portion, a base portion, a compliant membrane disposed substantially therebetween, two opposing edge sections, and an interface-facing surface, wherein the rigid body comprises a positioning feature arranged substantially orthogonal to a general plane of the interface-facing surface; and slider coupled to the two opposing edge sections and longitudinally articulable with respect to the rigid body.

(256) Concept 2. The pump cassette of concept 1 or any other concept, wherein the positioning feature is defined in part by the frame portion.

(257) Concept 3. The pump cassette of concept 1 or any other concept, wherein the positioning feature comprises a positioning port defining a recess into the interface-facing surface.

(258) Concept 4. The pump cassette of concept 1 or any other concept, wherein the positioning feature comprises a positioning protrusion extending from the interface-facing surface.

(259) Concept 5. The pump cassette of concept 4 or any other concept, wherein the rigid body further comprises a controllable fluid pathway defined in part by the compliant membrane and extending from an inlet port to an outlet port, and

(260) wherein the positioning protrusion comprises a fluid pathway extension member of the controllable fluid pathway.

(261) Concept 6. The pump cassette of concept 4 or any other concept, wherein the rigid body further comprises a controllable fluid pathway defined in part by the compliant membrane and extending from an inlet port to an outlet port, and wherein the positioning protrusion comprises a

pump structure configured to urge fluid through the controllable fluid pathway.

(262) Concept 7. The pump cassette of concept 6 or any other concept, wherein the pump structure comprises an external support wall portion extending from the interface-facing surface.

(263) Concept 8. The pump cassette of concept 1 or any other concept, wherein the positioning feature comprises a first positioning protrusion and a second positioning protrusion, each extending from the interface-facing surface.

(264) Concept 9. A pump cassette comprising: a rigid body comprising a compliant membrane, two opposing edge sections, and an interface-facing surface, wherein the rigid body comprises a controllable fluid pathway including a pump chamber, the controllable fluid pathway defined in part by the compliant membrane and extending from an inlet port to an outlet port; a pump assembly including a pump drive mechanism operably accessible from the interface-facing surface, wherein the pump drive mechanism is longitudinally articulable with respect to the rigid body; and a slider coupled to the two opposing edge sections and longitudinally articulable with respect to the rigid body.

(265) Concept 10. The pump cassette of concept 9 or any other concept, wherein the pump assembly further comprises a piston disposed within the rigid body and the pump drive mechanism is operably coupled to the piston.

(266) Concept 11. The pump cassette of concept 10 or any other concept, wherein the piston is arranged in a position above an opening of the pump chamber such that a piston head can vary a volume of the pump chamber when the pump drive mechanism is longitudinally articulated.

(267) Concept 12. The pump cassette of concept 11 or any other concept, wherein the position above the pump chamber is a position that is proximal to the inlet port and distal from the outlet port.

(268) Concept 13. The pump cassette of concept 10 or any other concept, wherein the piston is arranged in a position below an opening of the pump chamber such that a piston head can vary a volume of the pump chamber when the pump drive mechanism is longitudinally articulated.

(269) Concept 14. The pump cassette of concept 13 or any other concept, wherein the position below the pump chamber is a position that is proximal to the outlet port and distal from the inlet port.

(270) Concept 15. The pump cassette of concept 10 or any other concept, wherein the piston is arranged in a rail for guiding the piston during a piston stroke.

(271) Concept 16. The pump cassette of concept 9 or any other concept, wherein the rigid body comprises a positioning feature arranged on the interface-facing surface such that the positioning feature is proximal to the pump chamber and distal from either of the inlet port and the outlet port.

(272) Concept 17. The pump cassette of concept 9 or any other concept, wherein the slider comprises a lens area configured to magnify the controllable fluid pathway.

(273) Concept 18. An infusion pump system comprising: a processing unit; a cassette recess configured to receive a pump cassette, the cassette recess comprising (i) a plurality of cassette engagement slots configured for removably coupling the pump cassette, and (ii) a cassette-facing surface comprising a positioning feature slot extending orthogonal to a general plane of the cassette-facing surface.

(274) Concept 19. The infusion pump system of concept 18 or any other concept, wherein the cassette-facing surface comprises a positioning protrusion extending from the cassette-facing surface.

(275) Concept 20. The infusion pump system of concept 18 or any other concept, wherein the positioning feature slot comprises (i) a first positioning feature slot operably coupled to a pump actuator configured to apply a force orthogonal to the general plane of cassette-facing surface, and (ii) a second positioning feature slot operably coupled to a fluid sensor.

(276) The present disclosure is provided to enable any person skilled in the art to practice the various aspects described herein. The disclosure provides various examples of the subject

technology, and the subject technology is not limited to these examples. Various modifications to these aspects will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other aspects.

(277) One or more aspects or features of the subject matter described herein may be realized in digital electronic circuitry, integrated circuitry, specially designed ASICs (application specific integrated circuits), computer hardware, firmware, software, and/or combinations thereof. For example, infusion pump systems disclosed herein may include an electronic system with one or more processors embedded therein or coupled thereto. Such an electronic system may include various types of computer readable media and interfaces for various other types of computer readable media. Electronic system may include a bus, processing unit(s), a system memory, a read-only memory (ROM), a permanent storage device, an input device interface, an output device interface, and a network interface, for example.

(278) Bus may collectively represent all system, peripheral, and chipset buses that communicatively connect the numerous internal devices of electronic system of an infusion pump system. For instance, bus may communicatively connect processing unit(s) with ROM, system memory, and permanent storage device. From these various memory units, processing unit(s) may retrieve instructions to execute and data to process in order to execute various processes. The processing unit(s) can be a single processor or a multi-core processor in different implementations.

(279) A reference to an element in the singular is not intended to mean “one and only one” unless specifically so stated, but rather “one or more.” Unless specifically stated otherwise, the term “some” refers to one or more. Pronouns in the masculine (e.g., his) include the feminine and neuter gender (e.g., her and its) and vice versa. Headings and subheadings, if any, are used for convenience only and do not limit the invention.

(280) The word “exemplary” is used herein to mean “serving as an example or illustration.” Any aspect or design described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects or designs. In one aspect, various alternative configurations and operations described herein may be considered to be at least equivalent.

(281) As used herein, the phrase “at least one of” preceding a series of items, with the term “or” to separate any of the items, modifies the list as a whole, rather than each item of the list. The phrase “at least one of” does not require selection of at least one item; rather, the phrase allows a meaning that includes at least one of any one of the items, and/or at least one of any combination of the items, and/or at least one of each of the items. By way of example, the phrase “at least one of A, B, or C” may refer to: only A, only B, or only C; or any combination of A, B, and C.

(282) A phrase such as an “aspect” does not imply that such aspect is essential to the subject technology or that such aspect applies to all configurations of the subject technology. A disclosure relating to an aspect may apply to all configurations, or one or more configurations. An aspect may provide one or more examples. A phrase such as an aspect may refer to one or more aspects and vice versa. A phrase such as an “embodiment” does not imply that such embodiment is essential to the subject technology or that such embodiment applies to all configurations of the subject technology. A disclosure relating to an embodiment may apply to all embodiments, or one or more embodiments. An embodiment may provide one or more examples. A phrase such an embodiment may refer to one or more embodiments and vice versa. A phrase such as a “configuration” does not imply that such configuration is essential to the subject technology or that such configuration applies to all configurations of the subject technology. A disclosure relating to a configuration may apply to all configurations, or one or more configurations. A configuration may provide one or more examples. A phrase such a configuration may refer to one or more configurations and vice versa.

(283) In one aspect, unless otherwise stated, all measurements, values, ratings, positions, magnitudes, sizes, and other specifications that are set forth in this specification, including in the claims that follow, are approximate, not exact. In one aspect, they are intended to have a reasonable

range that is consistent with the functions to which they relate and with what is customary in the art to which they pertain.

(284) It is understood that the specific order or hierarchy of steps, or operations in the processes or methods disclosed are illustrations of exemplary approaches. Based upon implementation preferences or scenarios, it is understood that the specific order or hierarchy of steps, operations or processes may be rearranged. Some of the steps, operations or processes may be performed simultaneously. In some implementation preferences or scenarios, certain operations may or may not be performed. Some or all of the steps, operations, or processes may be performed automatically, without the intervention of a user. The accompanying method claims present elements of the various steps, operations or processes in a sample order, and are not meant to be limited to the specific order or hierarchy presented.

(285) All structural and functional equivalents to the elements of the various aspects described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the claims. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the claims. No claim element is to be construed under the provisions of 35 U.S.C. § 112 (f) unless the element is expressly recited using the phrase “means for” or, in the case of a method claim, the element is recited using the phrase “step for.” Furthermore, to the extent that the term “include,” “have,” or the like is used, such term is intended to be inclusive in a manner similar to the term “comprise” as “comprise” is interpreted when employed as a transitional word in a claim.

(286) The Title, Background, Summary, Brief Description of the Drawings and Abstract of the disclosure are hereby incorporated into the disclosure and are provided as illustrative examples of the disclosure, not as restrictive descriptions. It is submitted with the understanding that they will not be used to limit the scope or meaning of the claims. In addition, in the Detailed Description, it can be seen that the description provides illustrative examples and the various features are grouped together in various embodiments for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed subject matter requires more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter lies in less than all features of a single disclosed configuration or operation. The following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separately claimed subject matter.

(287) The claims are not intended to be limited to the aspects described herein, but is to be accorded the full scope consistent with the language claims and to encompass all legal equivalents. Notwithstanding, none of the claims are intended to embrace subject matter that fails to satisfy the requirement of 35 U.S.C. § 101, 102, or 103, nor should they be interpreted in such a way.

Claims

1. A pump system comprising: a processing unit; and a recess configured to receive a pump cartridge, the recess comprising: a plurality of cartridge engagement slots configured for removably coupling the pump cartridge; and a cartridge-facing surface comprising a positioning feature slot extending orthogonal to a general plane of the cartridge-facing surface, wherein the positioning feature slot comprises: a first positioning feature slot operably coupled to a pump actuator, wherein the pump actuator comprises a rotatable pin configured to rotate within the first positioning feature slot and engage with an elongate slot of a pump drive interface of the pump cartridge in a scotch-yoke configuration; and a second positioning feature slot operably coupled to a fluid sensor.
2. The pump system of claim 1, wherein the plurality of cartridge engagement slots comprises at least one L-shaped locking channel.
3. The pump system of claim 1, wherein the plurality of cartridge engagement slots comprises at

least one flat face ramp portion.

4. The pump system of claim 1, comprising an inlet recess formed through a portion of the recess.
 5. The pump system of claim 1, comprising an outlet recess formed through a portion of the recess.
 6. The pump system of claim 1, wherein the cartridge-facing surface comprises a positioning protrusion extending from the cartridge-facing surface.
 7. The pump system of claim 6, wherein the positioning protrusion is disposed proximal to an inlet-side valve actuator.
 8. The pump system of claim 6, wherein the positioning protrusion is disposed proximal to an outlet-side valve actuator.
 9. The pump system of claim 1, the recess comprising an upstream pressure sensing probe.
 10. The pump system of claim 1, the recess comprising a downstream pressure sensing probe.
 11. The pump system of claim 1, wherein the rotatable pin of the pump actuator is offset from a central axis of the pump actuator and configured to rotate around the central axis of the pump actuator.
 12. A pump system comprising: a processing unit; and a recess configured to receive a pump cartridge, the recess comprising: a cartridge-facing surface including a positioning feature slot extending orthogonal to a general plane of the cartridge-facing surface, wherein the positioning feature slot comprises: a first positioning feature slot operably coupled to a pump actuator, wherein the pump actuator comprises a rotatable pin configured to rotate within the first positioning feature slot and engage with an elongate slot of a pump drive interface of the pump cartridge in a scotch-yoke configuration; and a second positioning feature slot operably coupled to a fluid sensor.
 13. The pump system of claim 12, comprising an inlet recess formed through a portion of the cartridge recess.
 14. The pump system of claim 12, comprising an outlet recess formed through a portion of the cartridge recess.
 15. The pump system of claim 12, wherein the cartridge-facing surface comprises a positioning protrusion extending from the cartridge-facing surface.
 16. The pump system of claim 15, wherein the positioning protrusion is disposed proximal to an inlet-side valve actuator.
 17. The pump system of claim 15, wherein the positioning protrusion is disposed proximal to an outlet-side valve actuator.
 18. The pump system of claim 12, the cartridge recess comprising an upstream pressure sensing probe.
 19. The pump system of claim 12, the cartridge recess comprising a downstream pressure sensing probe.
 20. The pump system of claim 12, wherein the rotatable pin of the pump actuator is offset from a central axis of the pump actuator and configured to rotate around the central axis of the pump actuator.
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